



Danmarks Tekniske Universitet

Department of Civil Engineering

Multi-Sample Comparison Report

Hydraulic Conductivity Analysis - 15 Samples

Project: DTU Sustian

Date: June 23, 2026

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Report Date: June 23, 2026 at 20:20

Number of Samples: 15
Temperature: 20 °C
Porosity: Varies by sample (0.26 to 0.435)

Executive Summary

Comparison Analysis: This report compares 15 soil samples based on grain size distribution and hydraulic conductivity estimates.

Key Findings:

- Highest permeability: Sample_1_Odong (9.54e-04 m/s)
- Lowest permeability: Example_2_Case_1_Vukovic_15C (1.14e-06 m/s)

- Permeability range: 837.4x difference

Methodology

Comparative Analysis Approach

This comparison report presents a side-by-side analysis of multiple soil samples to identify patterns, variations, and relationships between grain size characteristics and hydraulic conductivity estimates.

Analysis Components

Grain Size Comparison: Overlapping distribution curves allow visual assessment of particle size variations between samples.

K-Value Comparison: Box plots and statistical summaries reveal the range and reliability of hydraulic conductivity estimates across samples.

Method Reliability: A reliability matrix shows which empirical methods are applicable for each sample, helping identify the most suitable estimation approaches.

Results & Analysis

Sample Overview

Sample	Temp (°C)	Porosity	D10 (mm)	D50 (mm)	D60 (mm)	Cu	Soil Type	K geometric mean (m/s)
Example_1_Case_1_Vukovic	20.00	0.384	0.150	0.485	0.550	3.67	Poorly-graded sand	2.73e-04
Example_1_Case_2_Vukovic_15C	20.00	0.384	0.150	0.480	0.550	3.67	Poorly-graded sand	2.74e-04
Example_2_Case_1_Vukovic_15C	20.00	0.26	0.014	0.210	0.300	21.43	Well-graded sand	1.14e-06
Sample_1_Odong	20.00	0.346	0.341	1.181	1.886	5.53	Poorly-graded sand	9.54e-04
Thomson_SERDP_Borden_Sand	20.00	0.407	0.070	0.180	0.195	2.79	Poorly-graded sand	5.01e-05
NBAL-01 [English]	20.00	0.435	0.171	0.295	0.319	1.86	Poorly-graded sand	3.64e-04
NBAL-02 [English]	20.00	0.399	0.255	0.655	0.785	3.08	Poorly-graded sand	7.45e-04
NBAL-03 [English]	20.00	0.405	0.122	0.305	0.350	2.86	Poorly-graded sand	1.94e-04
NBAL-04 [English]	20.00	0.35	N/A	0.173	0.205	N/A	Sand	8.50e-05
NBAL-05 [English]	20.00	0.419	0.098	0.192	0.233	2.38	Poorly-graded sand	1.21e-04
NBAL-06 [English]	20.00	0.421	0.257	0.526	0.592	2.30	Poorly-graded sand	8.28e-04
NBAL-07 [English]	20.00	0.425	0.219	0.425	0.473	2.16	Poorly-graded sand	6.05e-04
NBAL-08 [English]	20.00	0.423	0.125	0.244	0.281	2.25	Poorly-graded sand	1.94e-04
NBAL-09 [English]	20.00	0.432	0.164	0.297	0.325	1.97	Poorly-graded sand	3.38e-04

Sample	Temp (°C)	Porosity	D10 (mm)	D50 (mm)	D60 (mm)	Cu	Soil Type	K geometric mean (m/s)
NBAL-10 [English]	20.00	0.394	0.178	0.516	0.583	3.27	Poorly- graded sand	4.18e-04

Grain Parameters Comparison

Parameter	Sample	Value
D10 (mm)	Example_1_Case_1_Vukovic	0.150
D10 (mm)	Example_1_Case_2_Vukovic_15C	0.150
D10 (mm)	Example_2_Case_1_Vukovic_15C	0.014
D10 (mm)	Sample_1_Odong	0.341
D10 (mm)	Thomson_SERDP_Borden_Sand	0.070
D10 (mm)	NBAL-01 [English]	0.171
D10 (mm)	NBAL-02 [English]	0.255
D10 (mm)	NBAL-03 [English]	0.122
D10 (mm)	NBAL-04 [English]	-
D10 (mm)	NBAL-05 [English]	0.098
D10 (mm)	NBAL-06 [English]	0.257
D10 (mm)	NBAL-07 [English]	0.219
D10 (mm)	NBAL-08 [English]	0.125
D10 (mm)	NBAL-09 [English]	0.164
D10 (mm)	NBAL-10 [English]	0.178
D10 (mm) summary	Included samples	Mean: 0.165 Std: 0.081
D50 (mm)	Example_1_Case_1_Vukovic	0.485
D50 (mm)	Example_1_Case_2_Vukovic_15C	0.480
D50 (mm)	Example_2_Case_1_Vukovic_15C	0.210
D50 (mm)	Sample_1_Odong	1.181
D50 (mm)	Thomson_SERDP_Borden_Sand	0.180
D50 (mm)	NBAL-01 [English]	0.295
D50 (mm)	NBAL-02 [English]	0.655
D50 (mm)	NBAL-03 [English]	0.305
D50 (mm)	NBAL-04 [English]	0.173
D50 (mm)	NBAL-05 [English]	0.192
D50 (mm)	NBAL-06 [English]	0.526
D50 (mm)	NBAL-07 [English]	0.425
D50 (mm)	NBAL-08 [English]	0.244
D50 (mm)	NBAL-09 [English]	0.297
D50 (mm)	NBAL-10 [English]	0.516
D50 (mm) summary	Included samples	Mean: 0.411 Std: 0.252
D60 (mm)	Example_1_Case_1_Vukovic	0.550
D60 (mm)	Example_1_Case_2_Vukovic_15C	0.550
D60 (mm)	Example_2_Case_1_Vukovic_15C	0.300

Parameter	Sample	Value
D60 (mm)	Sample_1_Odong	1.886
D60 (mm)	Thomson_SERDP_Borden_Sand	0.195
D60 (mm)	NBAL-01 [English]	0.319
D60 (mm)	NBAL-02 [English]	0.785
D60 (mm)	NBAL-03 [English]	0.350
D60 (mm)	NBAL-04 [English]	0.205
D60 (mm)	NBAL-05 [English]	0.233
D60 (mm)	NBAL-06 [English]	0.592
D60 (mm)	NBAL-07 [English]	0.473
D60 (mm)	NBAL-08 [English]	0.281
D60 (mm)	NBAL-09 [English]	0.325
D60 (mm)	NBAL-10 [English]	0.583
D60 (mm) summary	Included samples	Mean: 0.508 Std: 0.404
Cu	Example_1_Case_1_Vukovic	3.67
Cu	Example_1_Case_2_Vukovic_15C	3.67
Cu	Example_2_Case_1_Vukovic_15C	21.43
Cu	Sample_1_Odong	5.53
Cu	Thomson_SERDP_Borden_Sand	2.79
Cu	NBAL-01 [English]	1.86
Cu	NBAL-02 [English]	3.08
Cu	NBAL-03 [English]	2.86
Cu	NBAL-04 [English]	-
Cu	NBAL-05 [English]	2.38
Cu	NBAL-06 [English]	2.30
Cu	NBAL-07 [English]	2.16
Cu	NBAL-08 [English]	2.25
Cu	NBAL-09 [English]	1.97
Cu	NBAL-10 [English]	3.27
Cu summary	Included samples	Mean: 4.23 Std: 4.86
Cc	Example_1_Case_1_Vukovic	1.48
Cc	Example_1_Case_2_Vukovic_15C	1.57
Cc	Example_2_Case_1_Vukovic_15C	1.50
Cc	Sample_1_Odong	0.54
Cc	Thomson_SERDP_Borden_Sand	1.56
Cc	NBAL-01 [English]	1.10
Cc	NBAL-02 [English]	0.84

Parameter	Sample	Value
Cc	NBAL-03 [English]	1.08
Cc	NBAL-04 [English]	-
Cc	NBAL-05 [English]	0.92
Cc	NBAL-06 [English]	1.03
Cc	NBAL-07 [English]	1.06
Cc	NBAL-08 [English]	0.88
Cc	NBAL-09 [English]	1.06
Cc	NBAL-10 [English]	1.40
Cc summary	Included samples	Mean: 1.15 Std: 0.30

K-Value Aggregate Summary

Scope	Datasets	K geometric mean (m/s)	K arithmetic mean (m/s)	K median (m/s)	ln(K) std dev	Included K cells	Warning cells	Permeability class
Overall	15	2.49e-04	4.88e-04	3.02e-04	1.525	188 / 240	52	High (Clean Sand)
test	3	3.98e-04	6.39e-04	3.44e-04	0.931	40 / 48	8	High (Clean Sand)
test2	2	1.15e-05	4.10e-05	3.41e-05	2.261	18 / 32	14	Moderate (Fine Sand)
test3	7	3.44e-04	5.53e-04	3.99e-04	1.060	89 / 112	23	High (Clean Sand)
test4	3	3.01e-04	3.97e-04	2.73e-04	0.716	41 / 48	7	High (Clean Sand)

K-Value Calculations by Dataset and Method

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
Example_1_Case_1_Vukovic	Hazen	3.00e-04	25.93	OK	Yes	OK
Example_1_Case_1_Vukovic	Hazen_1892	2.25e-04	19.44	Error	No	Empirical mm² variant (10°C reference)
Example_1_Case_1_Vukovic	Slichter	9.60e-05	8.29	OK	Yes	OK
Example_1_Case_1_Vukovic	Terzaghi	1.67e-04	14.43	OK	Yes	Reference 10°C calibration
Example_1_Case_1_Vukovic	Beyer	2.48e-04	21.43	OK	Yes	Within recommended range
Example_1_Case_1_Vukovic	Sauerbrei	3.39e-04	29.30	OK	Yes	OK
Example_1_Case_1_Vukovic	Kruger	5.16e-04	44.60	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
Example_1_Case_1_Vukovic	Kozeny-Carman	1.03e-03	89.22	OK	Yes	OK
Example_1_Case_1_Vukovic	Zunker	5.64e-04	48.73	OK	Yes	OK
Example_1_Case_1_Vukovic	Zamarin	6.65e-04	57.42	OK	Yes	OK
Example_1_Case_1_Vukovic	USBR	2.55e-04	22.04	OK	Yes	OK
Example_1_Case_1_Vukovic	Barr	1.34e-04	11.55	OK	Yes	OK
Example_1_Case_1_Vukovic	Alyamani-Sen	8.38e-05	7.24	OK	Yes	OK
Example_1_Case_1_Vukovic	Chapuis	1.74e-04	15.02	OK	Yes	OK
Example_1_Case_1_Vukovic	Shepherd	5.01e-04	43.27	OK	Yes	OK
Example_1_Case_1_Vukovic	Krumbein-Monk	3.22e-04	27.81	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Hazen	3.00e-04	25.93	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Hazen_1892	2.25e-04	19.44	Error	No	Empirical mm² variant (10°C reference)
Example_1_Case_2_Vukovic_15C	Slichter	9.60e-05	8.29	OK	Yes	OK

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
Example_1_Case_2_Vukovic_15C	Terzaghi	1.67e-04	14.43	OK	Yes	Reference 10°C calibration
Example_1_Case_2_Vukovic_15C	Beyer	2.48e-04	21.43	OK	Yes	Within recommended range
Example_1_Case_2_Vukovic_15C	Sauerbrei	3.39e-04	29.30	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Kruger	4.36e-04	37.67	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
Example_1_Case_2_Vukovic_15C	Kozeny-Carman	1.13e-03	97.65	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Zunker	5.65e-04	48.82	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Zamarin	6.08e-04	52.55	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	USBR	2.55e-04	22.04	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Barr	1.34e-04	11.55	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Alyamani-Sen	8.63e-05	7.46	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Chapuis	1.74e-04	15.02	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Shepherd	4.92e-04	42.54	OK	Yes	OK
Example_1_Case_2_Vukovic_15C	Krumbein-Monk	3.35e-04	28.96	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Hazen	1.16e-06	0.10	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
Example_2_Case_1_Vukovic_15C	Hazen_1892	1.96e-06	0.17	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm² variant (10°C reference)
Example_2_Case_1_Vukovic_15C	Slichter	2.32e-07	0.02	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Terzaghi	3.36e-07	0.03	OK	Yes	Reference 10°C calibration
Example_2_Case_1_Vukovic_15C	Beyer	1.38e-06	0.12	Warning	No	Outside stated U or D10 limits
Example_2_Case_1_Vukovic_15C	Sauerbrei	1.24e-06	0.11	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Kruger	1.14e-06	0.10	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
Example_2_Case_1_Vukovic_15C	Kozeny-Carman	8.17e-07	0.07	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Zunker	7.16e-07	0.06	Warning	No	Material contains fractions finer than 0.0025 mm
Example_2_Case_1_Vukovic_15C	Zamarin	1.00e-06	0.09	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	USBR	2.74e-06	0.24	Warning	No	U ≥ 5 exceeds USBR limit
Example_2_Case_1_Vukovic_15C	Barr	2.50e-07	0.02	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Alyamani-Sen	1.36e-05	1.18	Warning	No	Effective diameter non-positive
Example_2_Case_1_Vukovic_15C	Chapuis	4.13e-08	0.00	Warning	No	0.3 < n < 0.7 not satisfied; D10 > 0.10 mm not satisfied; Uniformity coefficient outside 2–12
Example_2_Case_1_Vukovic_15C	Shepherd	1.26e-04	10.87	OK	Yes	OK
Example_2_Case_1_Vukovic_15C	Krumbein-Monk	3.52e-06	0.30	Warning	No	sigma_phi = 3.07 suggests non-lognormal distribution
Sample_1_Odong	Hazen	1.29e-03	111.58	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
Sample_1_Odong	Hazen_1892	1.16e-03	100.62	Error	No	Empirical mm² variant (10°C reference)
Sample_1_Odong	Slichter	3.54e-04	30.55	OK	Yes	OK
Sample_1_Odong	Terzaghi	6.02e-04	52.03	OK	Yes	Reference 10°C calibration
Sample_1_Odong	Beyer	1.18e-03	101.69	OK	Yes	Within recommended range
Sample_1_Odong	Sauerbrei	6.79e-04	58.67	OK	Yes	OK
Sample_1_Odong	Kruger	1.98e-03	171.23	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
Sample_1_Odong	Kozeny-Carman	3.86e-03	333.21	OK	Yes	OK
Sample_1_Odong	Zunker	2.20e-03	189.70	OK	Yes	OK
Sample_1_Odong	Zamarin	2.56e-03	221.02	OK	Yes	OK

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
Sample_1_Odong	USBR	7.89e-04	68.18	Warning	No	U ≥ 5 exceeds USBR limit
Sample_1_Odong	Barr	4.51e-04	38.94	OK	Yes	OK
Sample_1_Odong	Alyamani-Sen	3.49e-04	30.19	OK	Yes	OK
Sample_1_Odong	Chapuis	7.59e-04	65.61	OK	Yes	OK
Sample_1_Odong	Shepherd	2.17e-03	187.83	OK	Yes	OK
Sample_1_Odong	Krumbein-Monk	5.93e-04	51.26	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Hazen	7.20e-05	6.22	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
Thomson_SERDP_Borden_Sand	Hazen_1892	4.90e-05	4.23	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm² variant (10°C reference)
Thomson_SERDP_Borden_Sand	Slichter	2.53e-05	2.18	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Terzaghi	4.43e-05	3.83	OK	Yes	Reference 10°C calibration
Thomson_SERDP_Borden_Sand	Beyer	5.70e-05	4.93	OK	Yes	Within recommended range
Thomson_SERDP_Borden_Sand	Sauerbrei	6.99e-05	6.04	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Kruger	4.06e-05	3.51	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
Thomson_SERDP_Borden_Sand	Kozeny-Carman	9.96e-05	8.60	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Zunker	4.98e-05	4.30	Warning	No	Material contains fractions finer than 0.0025 mm
Thomson_SERDP_Borden_Sand	Zamarin	5.44e-05	4.70	OK	Yes	OK
Thomson_SERDP_Borden_Sand	USBR	2.86e-05	2.47	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Barr	3.73e-05	3.23	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Alyamani-Sen	3.09e-05	2.67	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Chapuis	4.57e-05	3.95	Warning	No	D10 > 0.10 mm not satisfied

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
Thomson_SERDP_Borden_Sand	Shepherd	9.74e-05	8.42	OK	Yes	OK
Thomson_SERDP_Borden_Sand	Krumbein-Monk	6.41e-05	5.54	OK	Yes	OK
NBAL-01 [English]	Hazen	4.79e-04	41.42	OK	Yes	OK
NBAL-01 [English]	Hazen_1892	2.92e-04	25.26	Error	No	Empirical mm² variant (10°C reference)
NBAL-01 [English]	Slichter	1.88e-04	16.28	OK	Yes	OK
NBAL-01 [English]	Terzaghi	3.32e-04	28.72	OK	Yes	Reference 10°C calibration
NBAL-01 [English]	Beyer	3.67e-04	31.68	OK	Yes	Within recommended range
NBAL-01 [English]	Sauerbrei	3.99e-04	34.45	OK	Yes	OK
NBAL-01 [English]	Kruger	4.48e-04	38.71	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
NBAL-01 [English]	Kozeny-Carman	1.38e-03	119.24	OK	Yes	OK
NBAL-01 [English]	Zunker	6.63e-04	57.29	OK	Yes	OK
NBAL-01 [English]	Zamarin	6.38e-04	55.15	OK	Yes	OK
NBAL-01 [English]	USBR	1.32e-04	11.39	OK	Yes	OK
NBAL-01 [English]	Barr	3.01e-04	26.05	OK	Yes	OK
NBAL-01 [English]	Alyamani-Sen	3.08e-04	26.63	OK	Yes	OK
NBAL-01 [English]	Chapuis	5.41e-04	46.74	Warning	No	Uniformity coefficient outside 2–12
NBAL-01 [English]	Shepherd	2.20e-04	19.03	OK	Yes	OK
NBAL-01 [English]	Krumbein-Monk	2.88e-04	24.89	OK	Yes	OK
NBAL-02 [English]	Hazen	9.27e-04	80.10	OK	Yes	OK
NBAL-02 [English]	Hazen_1892	6.52e-04	56.29	Error	No	Empirical mm² variant (10°C reference)

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
NBAL-02 [English]	Slichter	3.15e-04	27.23	OK	Yes	OK
NBAL-02 [English]	Terzaghi	5.51e-04	47.63	OK	Yes	Reference 10°C calibration
NBAL-02 [English]	Beyer	7.44e-04	64.28	OK	Yes	Within recommended range
NBAL-02 [English]	Sauerbrei	6.21e-04	53.66	OK	Yes	OK
NBAL-02 [English]	Kruger	1.25e-03	108.05	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
NBAL-02 [English]	Kozeny-Carman	2.70e-03	233.49	OK	Yes	OK
NBAL-02 [English]	Zunker	1.66e-03	143.27	OK	Yes	OK
NBAL-02 [English]	Zamarin	1.75e-03	151.25	OK	Yes	OK
NBAL-02 [English]	USBR	3.46e-04	29.93	OK	Yes	OK
NBAL-02 [English]	Barr	4.56e-04	39.43	OK	Yes	OK
NBAL-02 [English]	Alyamani-Sen	4.11e-04	35.54	OK	Yes	OK
NBAL-02 [English]	Chapuis	7.38e-04	63.73	OK	Yes	OK
NBAL-02 [English]	Shepherd	8.22e-04	71.01	OK	Yes	OK
NBAL-02 [English]	Krumbein-Monk	7.05e-04	60.95	OK	Yes	OK
NBAL-03 [English]	Hazen	2.18e-04	18.87	OK	Yes	OK
NBAL-03 [English]	Hazen_1892	1.50e-04	12.94	Error	No	Empirical mm² variant (10°C reference)
NBAL-03 [English]	Slichter	7.61e-05	6.57	OK	Yes	OK
NBAL-03 [English]	Terzaghi	1.33e-04	11.51	OK	Yes	Reference 10°C calibration
NBAL-03 [English]	Beyer	1.74e-04	14.99	OK	Yes	Within recommended range
NBAL-03 [English]	Sauerbrei	1.78e-04	15.37	OK	Yes	OK

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
NBAL-03 [English]	Kruger	4.42e-04	38.19	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
NBAL-03 [English]	Kozeny-Carman	6.50e-04	56.13	OK	Yes	OK
NBAL-03 [English]	Zunker	6.06e-04	52.38	OK	Yes	OK
NBAL-03 [English]	Zamarin	6.26e-04	54.13	OK	Yes	OK
NBAL-03 [English]	USBR	8.10e-05	7.00	OK	Yes	OK
NBAL-03 [English]	Barr	1.12e-04	9.66	OK	Yes	OK
NBAL-03 [English]	Alyamani-Sen	9.96e-05	8.61	OK	Yes	OK
NBAL-03 [English]	Chapuis	1.54e-04	13.33	OK	Yes	OK
NBAL-03 [English]	Shepherd	2.33e-04	20.10	OK	Yes	OK
NBAL-03 [English]	Krumbein-Monk	1.98e-04	17.12	OK	Yes	OK
NBAL-04 [English]	Hazen	N/A	N/A	Warning	No	D10 required
NBAL-04 [English]	Hazen_1892	N/A	N/A	Warning	No	D10 required
NBAL-04 [English]	Slichter	N/A	N/A	Warning	No	D10 required
NBAL-04 [English]	Terzaghi	N/A	N/A	Warning	No	D10 required
NBAL-04 [English]	Beyer	N/A	N/A	Warning	No	D10 and D60 required
NBAL-04 [English]	Sauerbrei	1.58e-05	1.36	OK	Yes	OK
NBAL-04 [English]	Kruger	2.07e-04	17.85	OK	Yes	OK
NBAL-04 [English]	Kozeny-Carman	1.02e-04	8.82	OK	Yes	OK
NBAL-04 [English]	Zunker	2.47e-04	21.33	OK	Yes	OK
NBAL-04 [English]	Zamarin	2.77e-04	23.94	OK	Yes	OK
NBAL-04 [English]	USBR	8.32e-06	0.72	OK	Yes	OK
NBAL-04 [English]	Barr	N/A	N/A	Warning	No	D10 required

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
NBAL-04 [English]	Alyamani-Sen	N/A	N/A	Warning	No	D10 and D50 required
NBAL-04 [English]	Chapuis	N/A	N/A	Warning	No	D10 required
NBAL-04 [English]	Shepherd	9.12e-05	7.88	OK	Yes	OK
NBAL-04 [English]	Krumbein-Monk	1.58e-04	13.67	OK	Yes	OK
NBAL-05 [English]	Hazen	1.48e-04	12.80	Warning	No	Best for D10: 0.01-0.3 cm, UC < 5
NBAL-05 [English]	Hazen_1892	9.61e-05	8.30	Warning	No	D10 outside 0.01–0.3 cm; Empirical mm² variant (10°C reference)
NBAL-05 [English]	Slichter	5.46e-05	4.71	OK	Yes	OK
NBAL-05 [English]	Terzaghi	9.59e-05	8.29	OK	Yes	Reference 10°C calibration
NBAL-05 [English]	Beyer	1.15e-04	9.96	OK	Yes	Within recommended range
NBAL-05 [English]	Sauerbrei	1.16e-04	10.02	OK	Yes	OK
NBAL-05 [English]	Kruger	1.97e-04	17.01	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
NBAL-05 [English]	Kozeny-Carman	4.67e-04	40.37	OK	Yes	OK
NBAL-05 [English]	Zunker	2.81e-04	24.29	OK	Yes	OK
NBAL-05 [English]	Zamarin	2.81e-04	24.28	OK	Yes	OK
NBAL-05 [English]	USBR	3.97e-05	3.43	OK	Yes	OK
NBAL-05 [English]	Barr	8.33e-05	7.20	OK	Yes	OK
NBAL-05 [English]	Alyamani-Sen	8.91e-05	7.70	OK	Yes	OK
NBAL-05 [English]	Chapuis	1.22e-04	10.52	Warning	No	D10 > 0.10 mm not satisfied
NBAL-05 [English]	Shepherd	1.08e-04	9.35	OK	Yes	OK
NBAL-05 [English]	Krumbein-Monk	1.16e-04	10.04	OK	Yes	OK

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
NBAL-06 [English]	Hazen	1.03e-03	88.69	OK	Yes	OK
NBAL-06 [English]	Hazen_1892	6.60e-04	57.02	Error	No	Empirical mm² variant (10°C reference)
NBAL-06 [English]	Slichter	3.82e-04	32.97	OK	Yes	OK
NBAL-06 [English]	Terzaghi	6.71e-04	57.99	OK	Yes	Reference 10°C calibration
NBAL-06 [English]	Beyer	7.96e-04	68.81	OK	Yes	Within recommended range
NBAL-06 [English]	Sauerbrei	8.20e-04	70.84	OK	Yes	OK
NBAL-06 [English]	Kruger	1.26e-03	108.54	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
NBAL-06 [English]	Kozeny-Carman	3.39e-03	292.61	OK	Yes	OK
NBAL-06 [English]	Zunker	1.76e-03	151.98	OK	Yes	OK
NBAL-06 [English]	Zamarin	1.77e-03	153.08	OK	Yes	OK
NBAL-06 [English]	USBR	3.66e-04	31.65	OK	Yes	OK
NBAL-06 [English]	Barr	5.87e-04	50.68	OK	Yes	OK
NBAL-06 [English]	Alyamani-Sen	5.80e-04	50.08	OK	Yes	OK
NBAL-06 [English]	Chapuis	1.03e-03	89.07	OK	Yes	OK
NBAL-06 [English]	Shepherd	5.73e-04	49.52	OK	Yes	OK
NBAL-06 [English]	Krumbein-Monk	5.38e-04	46.51	OK	Yes	OK
NBAL-07 [English]	Hazen	7.57e-04	65.42	OK	Yes	OK
NBAL-07 [English]	Hazen_1892	4.79e-04	41.35	Error	No	Empirical mm² variant (10°C reference)
NBAL-07 [English]	Slichter	2.87e-04	24.76	OK	Yes	OK
NBAL-07 [English]	Terzaghi	5.04e-04	43.59	OK	Yes	Reference 10°C calibration

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
NBAL-07 [English]	Beyer	5.84e-04	50.50	OK	Yes	Within recommended range
NBAL-07 [English]	Sauerbrei	6.38e-04	55.12	OK	Yes	OK
NBAL-07 [English]	Kruger	7.60e-04	65.64	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
NBAL-07 [English]	Kozeny-Carman	2.20e-03	189.80	OK	Yes	OK
NBAL-07 [English]	Zunker	1.08e-03	93.39	OK	Yes	OK
NBAL-07 [English]	Zamarin	1.07e-03	92.84	OK	Yes	OK
NBAL-07 [English]	USBR	2.62e-04	22.67	OK	Yes	OK
NBAL-07 [English]	Barr	4.46e-04	38.53	OK	Yes	OK
NBAL-07 [English]	Alyamani-Sen	4.47e-04	38.59	OK	Yes	OK
NBAL-07 [English]	Chapuis	7.80e-04	67.39	OK	Yes	OK
NBAL-07 [English]	Shepherd	4.03e-04	34.85	OK	Yes	OK
NBAL-07 [English]	Krumbein-Monk	5.09e-04	44.01	OK	Yes	OK
NBAL-08 [English]	Hazen	2.45e-04	21.20	OK	Yes	OK
NBAL-08 [English]	Hazen_1892	1.57e-04	13.54	Error	No	Empirical mm² variant (10°C reference)
NBAL-08 [English]	Slichter	9.18e-05	7.93	OK	Yes	OK
NBAL-08 [English]	Terzaghi	1.62e-04	13.96	OK	Yes	Reference 10°C calibration
NBAL-08 [English]	Beyer	1.90e-04	16.41	OK	Yes	Within recommended range
NBAL-08 [English]	Sauerbrei	1.81e-04	15.68	OK	Yes	OK
NBAL-08 [English]	Kruger	2.73e-04	23.63	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
NBAL-08 [English]	Kozeny-Carman	7.28e-04	62.92	OK	Yes	OK

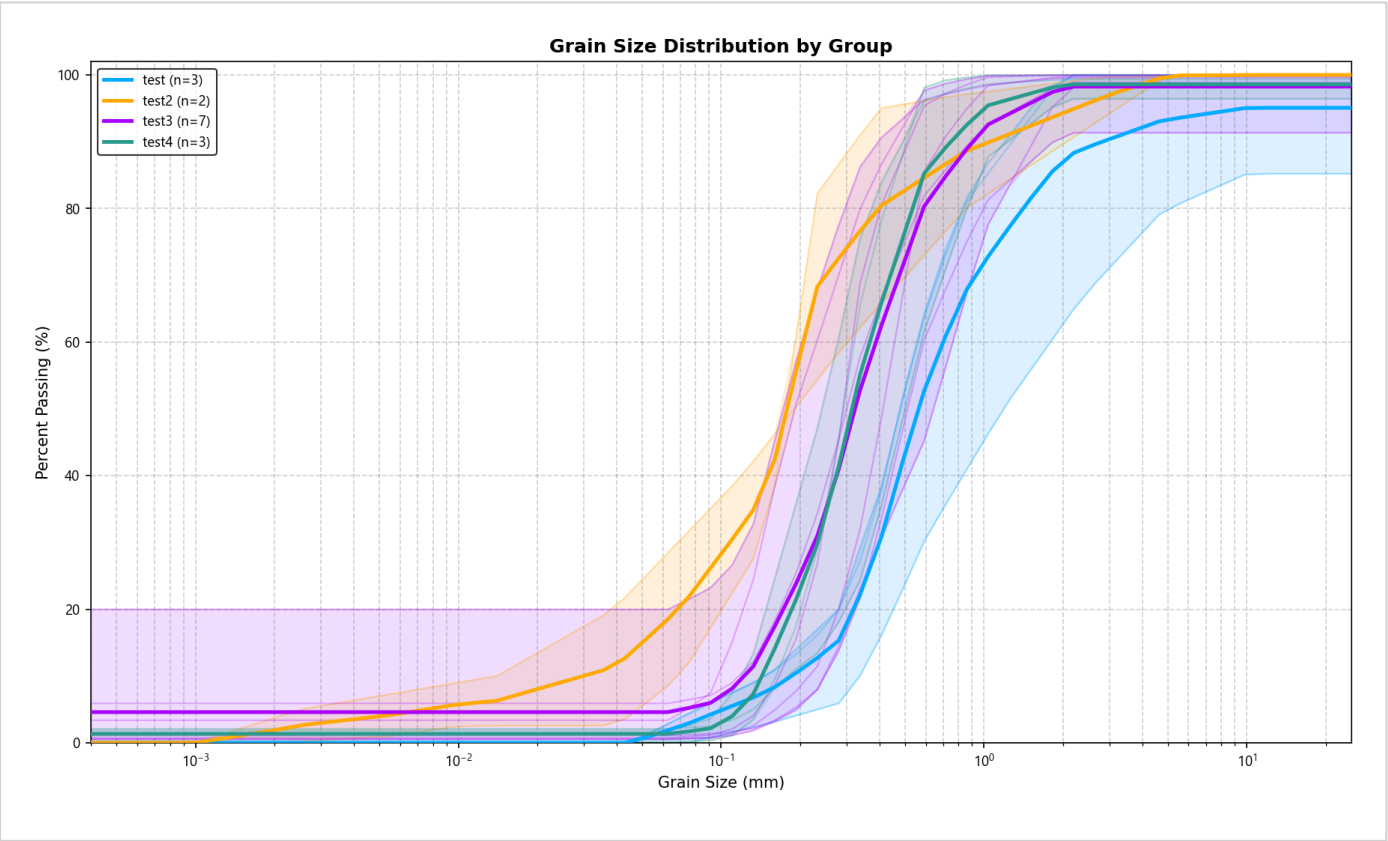
Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
NBAL-08 [English]	Zunker	3.94e-04	34.03	OK	Yes	OK
NBAL-08 [English]	Zamarin	3.90e-04	33.73	OK	Yes	OK
NBAL-08 [English]	USBR	6.11e-05	5.28	OK	Yes	OK
NBAL-08 [English]	Barr	1.42e-04	12.26	OK	Yes	OK
NBAL-08 [English]	Alyamani-Sen	1.46e-04	12.59	OK	Yes	OK
NBAL-08 [English]	Chapuis	2.23e-04	19.29	OK	Yes	OK
NBAL-08 [English]	Shepherd	1.61e-04	13.94	OK	Yes	OK
NBAL-08 [English]	Krumbein-Monk	1.69e-04	14.60	OK	Yes	OK
NBAL-09 [English]	Hazen	4.37e-04	37.79	OK	Yes	OK
NBAL-09 [English]	Hazen_1892	2.70e-04	23.35	Error	No	Empirical mm² variant (10°C reference)
NBAL-09 [English]	Slichter	1.69e-04	14.64	OK	Yes	OK
NBAL-09 [English]	Terzaghi	2.99e-04	25.81	OK	Yes	Reference 10°C calibration
NBAL-09 [English]	Beyer	3.36e-04	28.99	OK	Yes	Within recommended range
NBAL-09 [English]	Sauerbrei	3.58e-04	30.95	OK	Yes	OK
NBAL-09 [English]	Kruger	4.24e-04	36.66	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
NBAL-09 [English]	Kozeny-Carman	1.38e-03	119.50	OK	Yes	OK
NBAL-09 [English]	Zunker	6.22e-04	53.71	OK	Yes	OK
NBAL-09 [English]	Zamarin	6.04e-04	52.22	OK	Yes	OK
NBAL-09 [English]	USBR	1.22e-04	10.56	OK	Yes	OK
NBAL-09 [English]	Barr	2.68e-04	23.18	OK	Yes	OK

Sample	Method	K (m/s)	K (m/d)	Status	Included in Mean	Notes
NBAL-09 [English]	Alyamani-Sen	2.73e-04	23.57	OK	Yes	OK
NBAL-09 [English]	Chapuis	4.68e-04	40.40	Warning	No	Uniformity coefficient outside 2–12
NBAL-09 [English]	Shepherd	2.22e-04	19.22	OK	Yes	OK
NBAL-09 [English]	Krumbein-Monk	2.72e-04	23.53	OK	Yes	OK
NBAL-10 [English]	Hazen	4.43e-04	38.25	OK	Yes	OK
NBAL-10 [English]	Hazen_1892	3.18e-04	27.47	Error	No	Empirical mm² variant (10°C reference)
NBAL-10 [English]	Slichter	1.47e-04	12.73	OK	Yes	OK
NBAL-10 [English]	Terzaghi	2.57e-04	22.23	OK	Yes	Reference 10°C calibration
NBAL-10 [English]	Beyer	3.59e-04	30.99	OK	Yes	Within recommended range
NBAL-10 [English]	Sauerbrei	4.54e-04	39.21	OK	Yes	OK
NBAL-10 [English]	Kruger	8.66e-04	74.83	Warning	No	Best for medium sand (0.25 < D50 < 0.5 mm, UC > 5)
NBAL-10 [English]	Kozeny-Carman	1.60e-03	137.91	OK	Yes	OK
NBAL-10 [English]	Zunker	1.14e-03	98.19	OK	Yes	OK
NBAL-10 [English]	Zamarin	1.21e-03	104.53	OK	Yes	OK
NBAL-10 [English]	USBR	3.01e-04	26.01	OK	Yes	OK
NBAL-10 [English]	Barr	2.11e-04	18.19	OK	Yes	OK
NBAL-10 [English]	Alyamani-Sen	1.57e-04	13.61	OK	Yes	OK
NBAL-10 [English]	Chapuis	3.02e-04	26.10	OK	Yes	OK
NBAL-10 [English]	Shepherd	5.55e-04	47.94	OK	Yes	OK
NBAL-10 [English]	Krumbein-Monk	4.94e-04	42.64	OK	Yes	OK

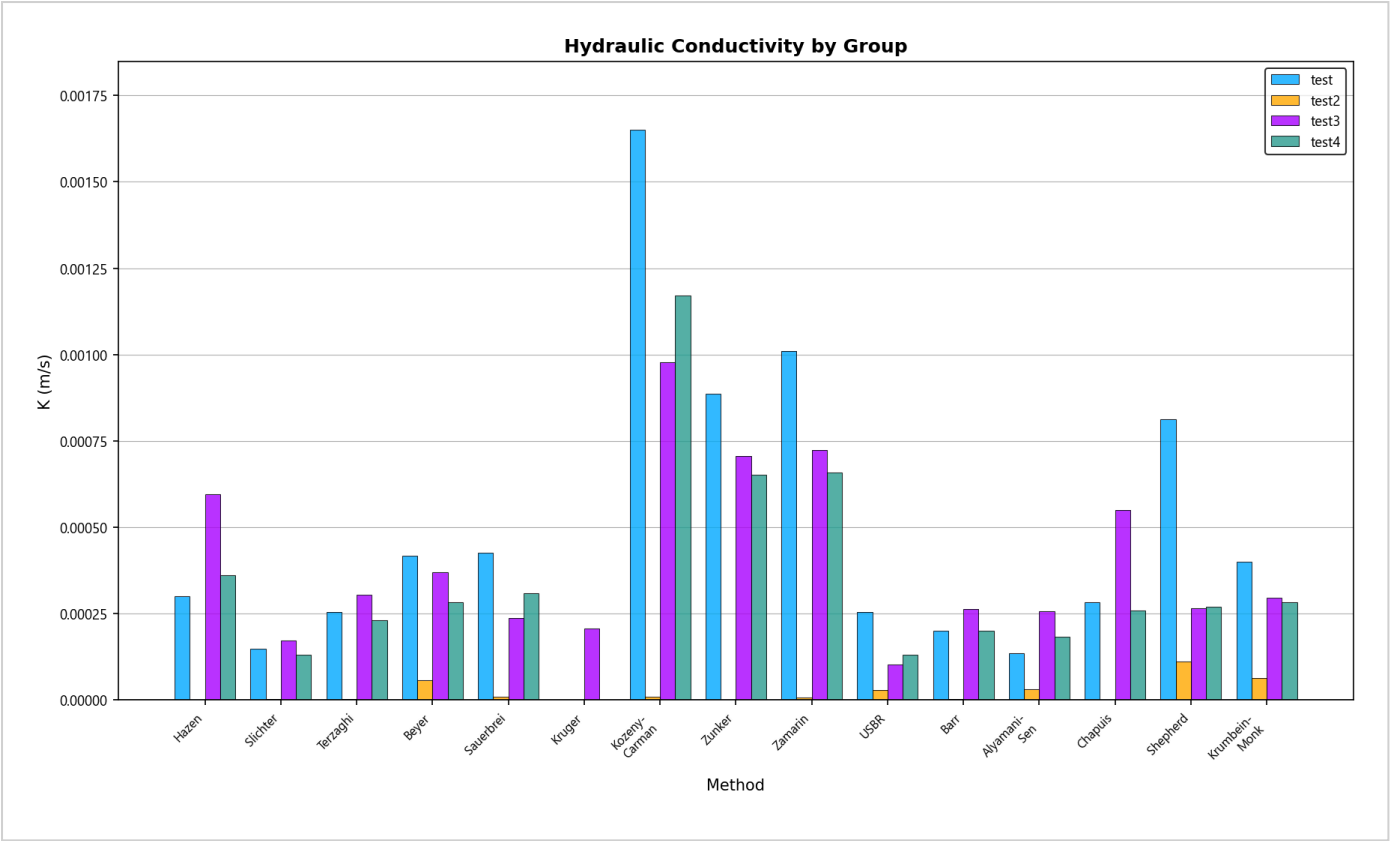
Permeability Classification Summary

Sample Name	K geometric mean (m/s)	Classification
Example_1_Case_1_Vukovic	2.73e-04	High (Clean Sand)
Example_1_Case_2_Vukovic_15C	2.74e-04	High (Clean Sand)
Example_2_Case_1_Vukovic_15C	1.14e-06	Low (Silt)
Sample_1_Odong	9.54e-04	High (Clean Sand)
Thomson_SERDP_Borden_Sand	5.01e-05	Moderate (Fine Sand)
NBAL-01 [English]	3.64e-04	High (Clean Sand)
NBAL-02 [English]	7.45e-04	High (Clean Sand)
NBAL-03 [English]	1.94e-04	High (Clean Sand)
NBAL-04 [English]	8.50e-05	Moderate (Fine Sand)
NBAL-05 [English]	1.21e-04	High (Clean Sand)
NBAL-06 [English]	8.28e-04	High (Clean Sand)
NBAL-07 [English]	6.05e-04	High (Clean Sand)
NBAL-08 [English]	1.94e-04	High (Clean Sand)
NBAL-09 [English]	3.38e-04	High (Clean Sand)
NBAL-10 [English]	4.18e-04	High (Clean Sand)

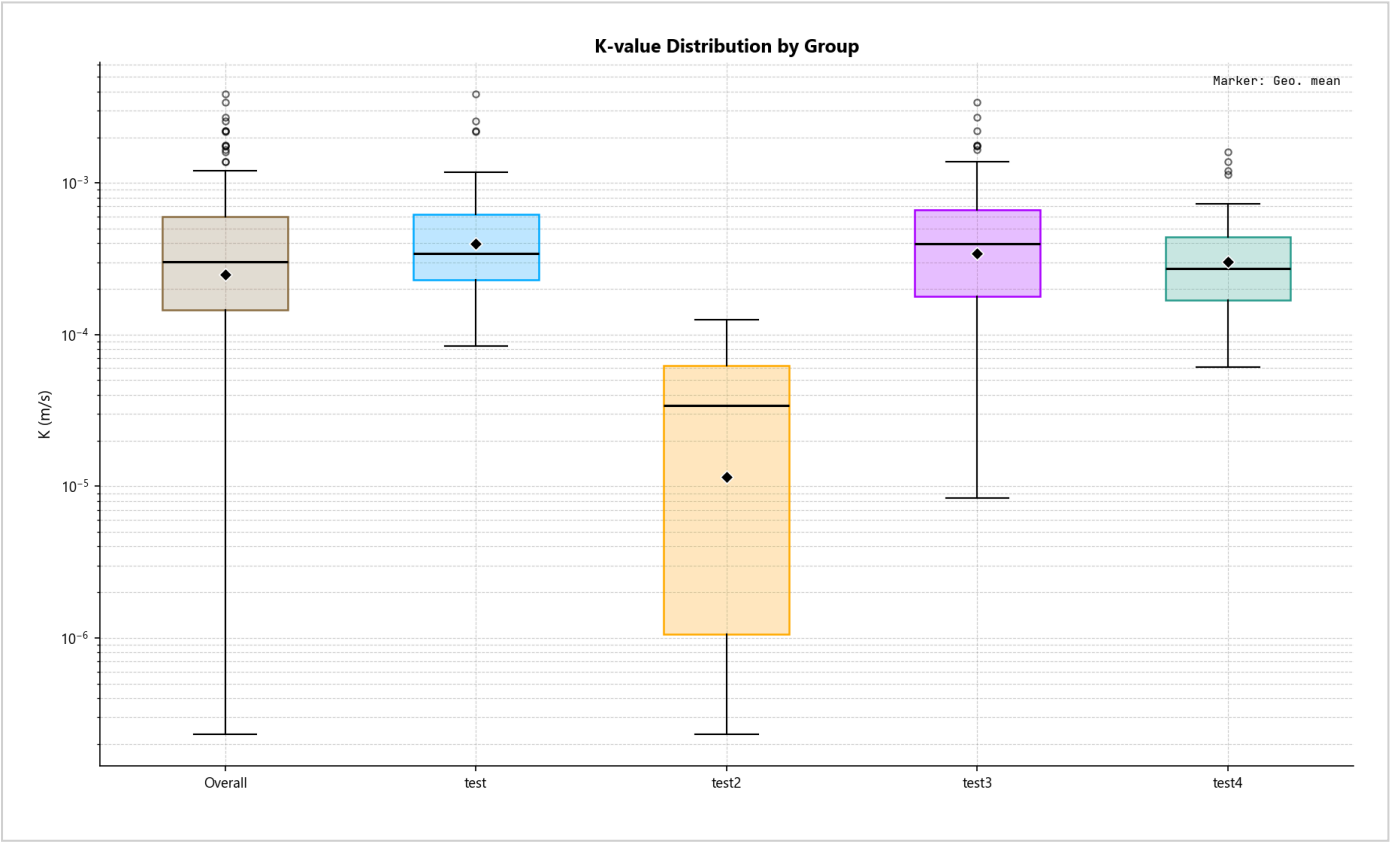
Grain Size Distribution Comparison



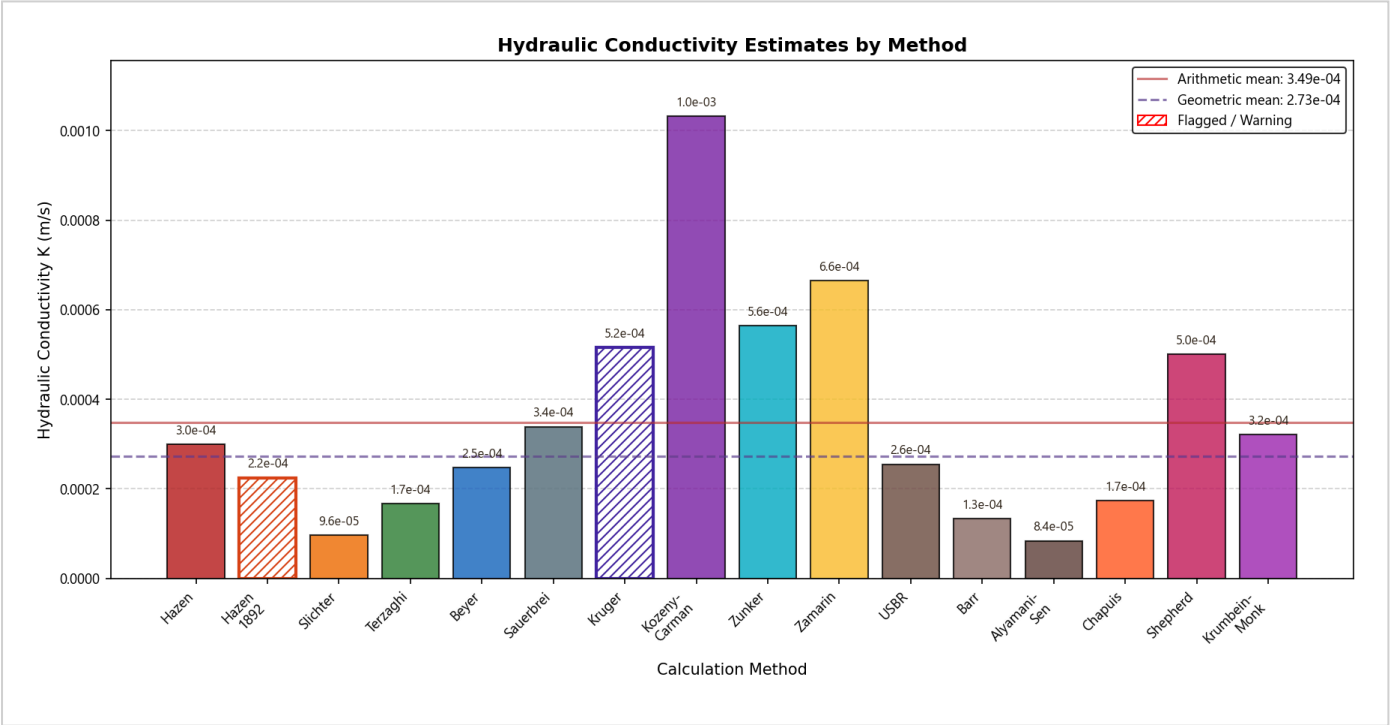
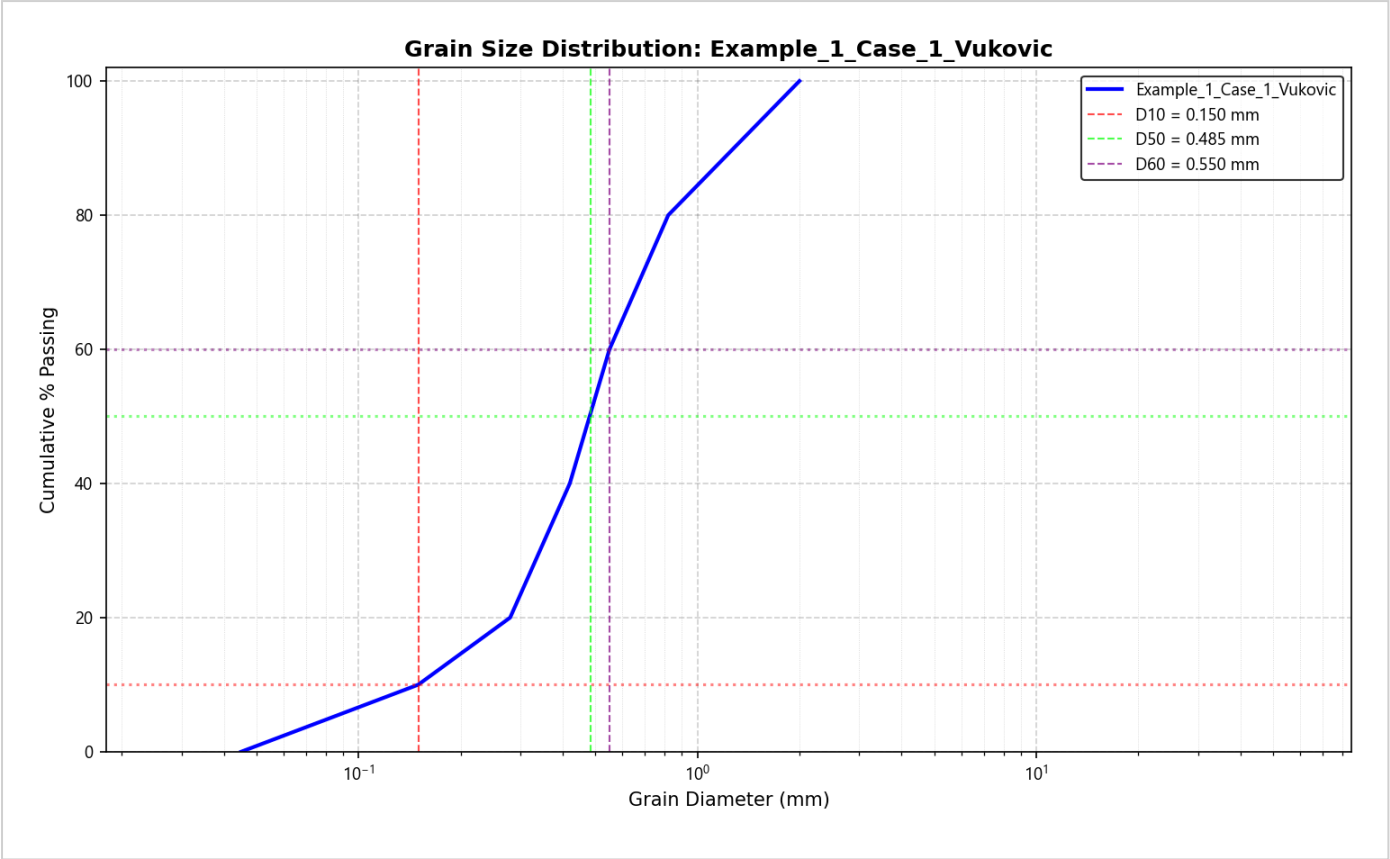
Hydraulic Conductivity by Method

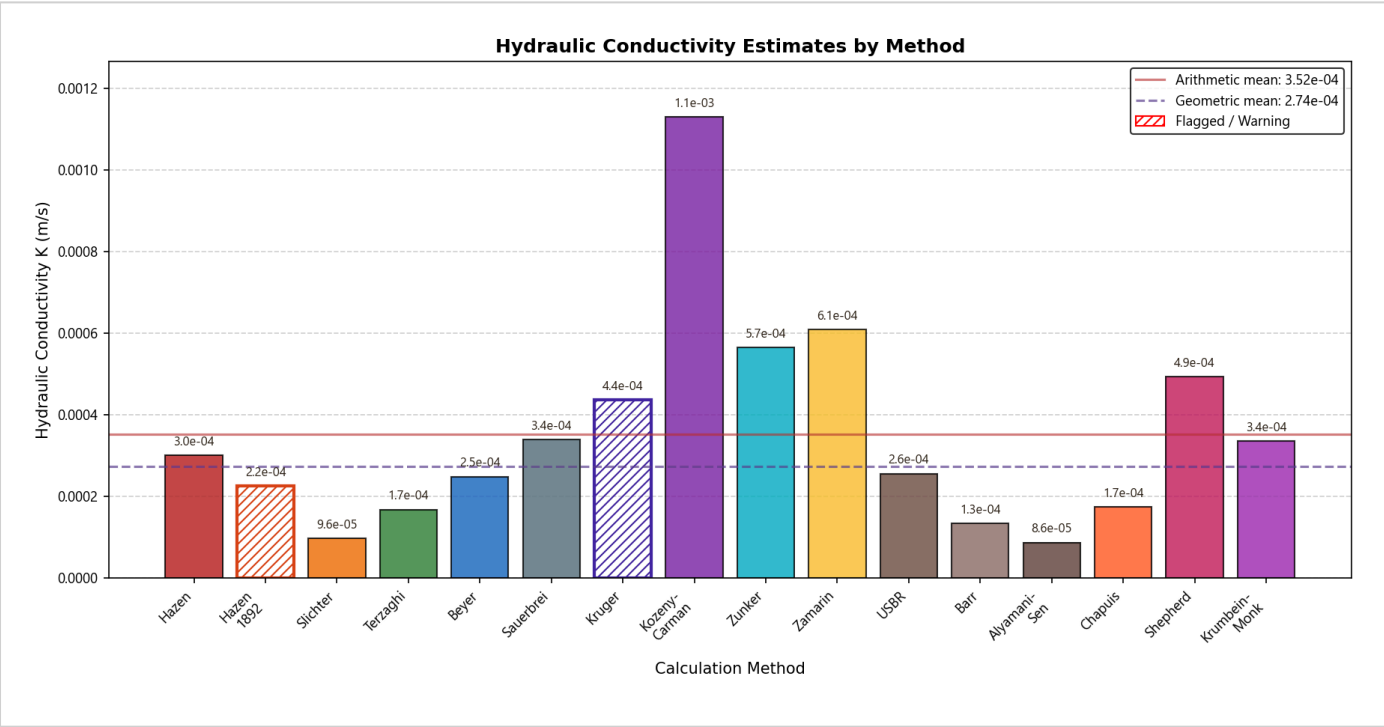
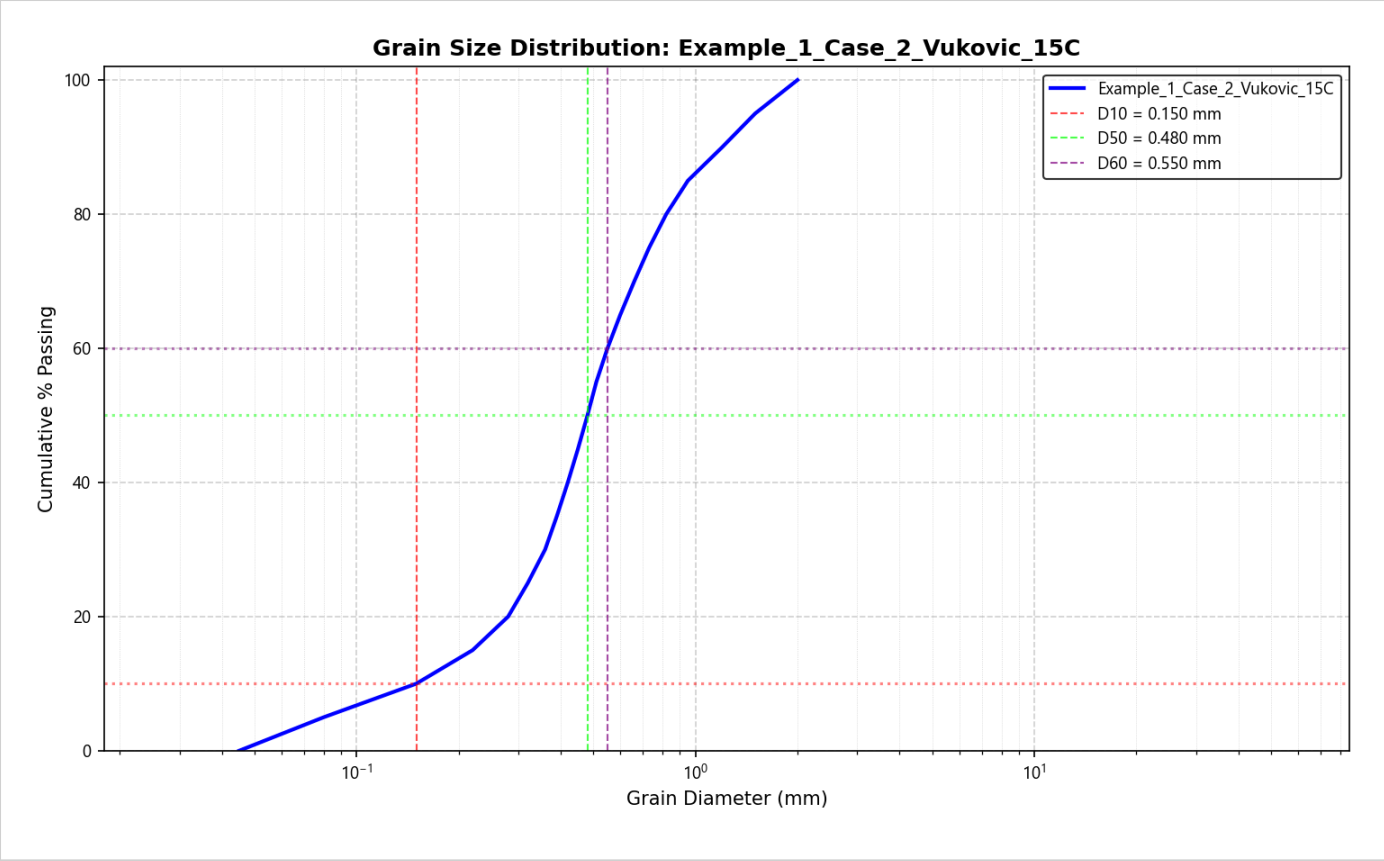


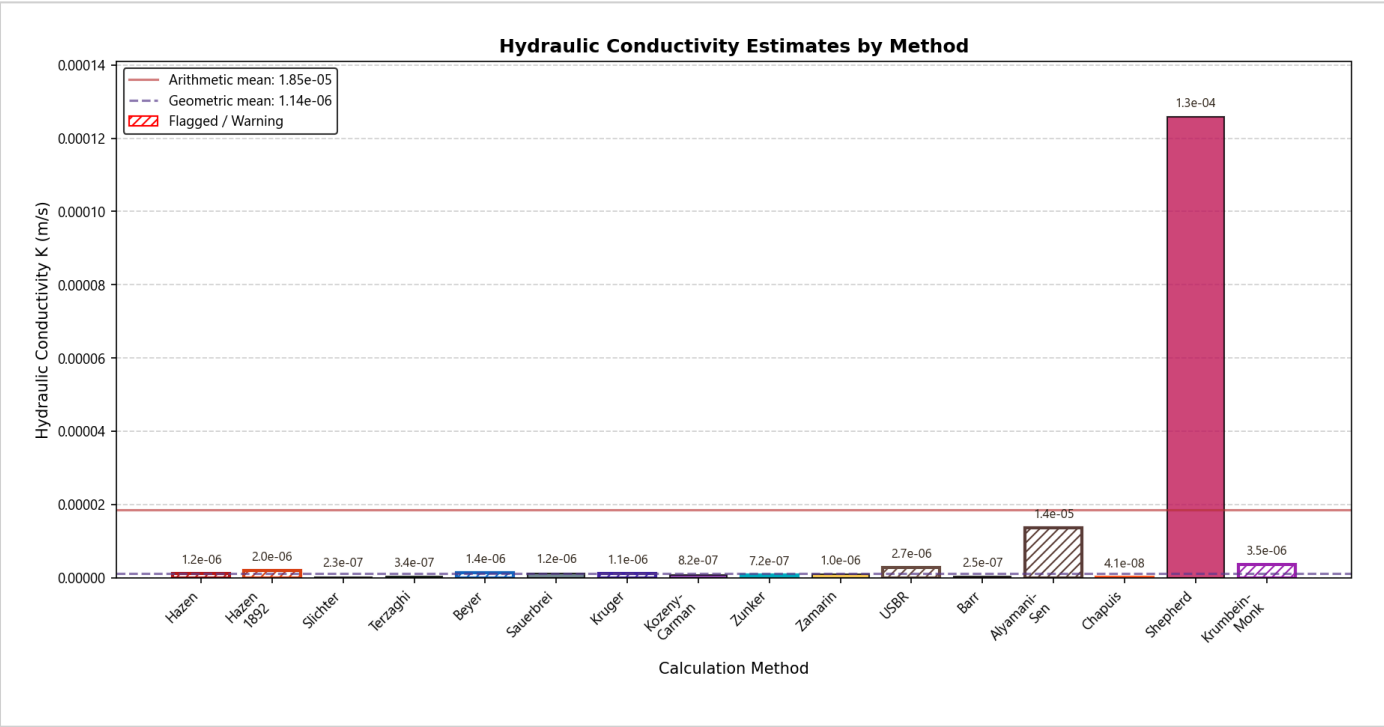
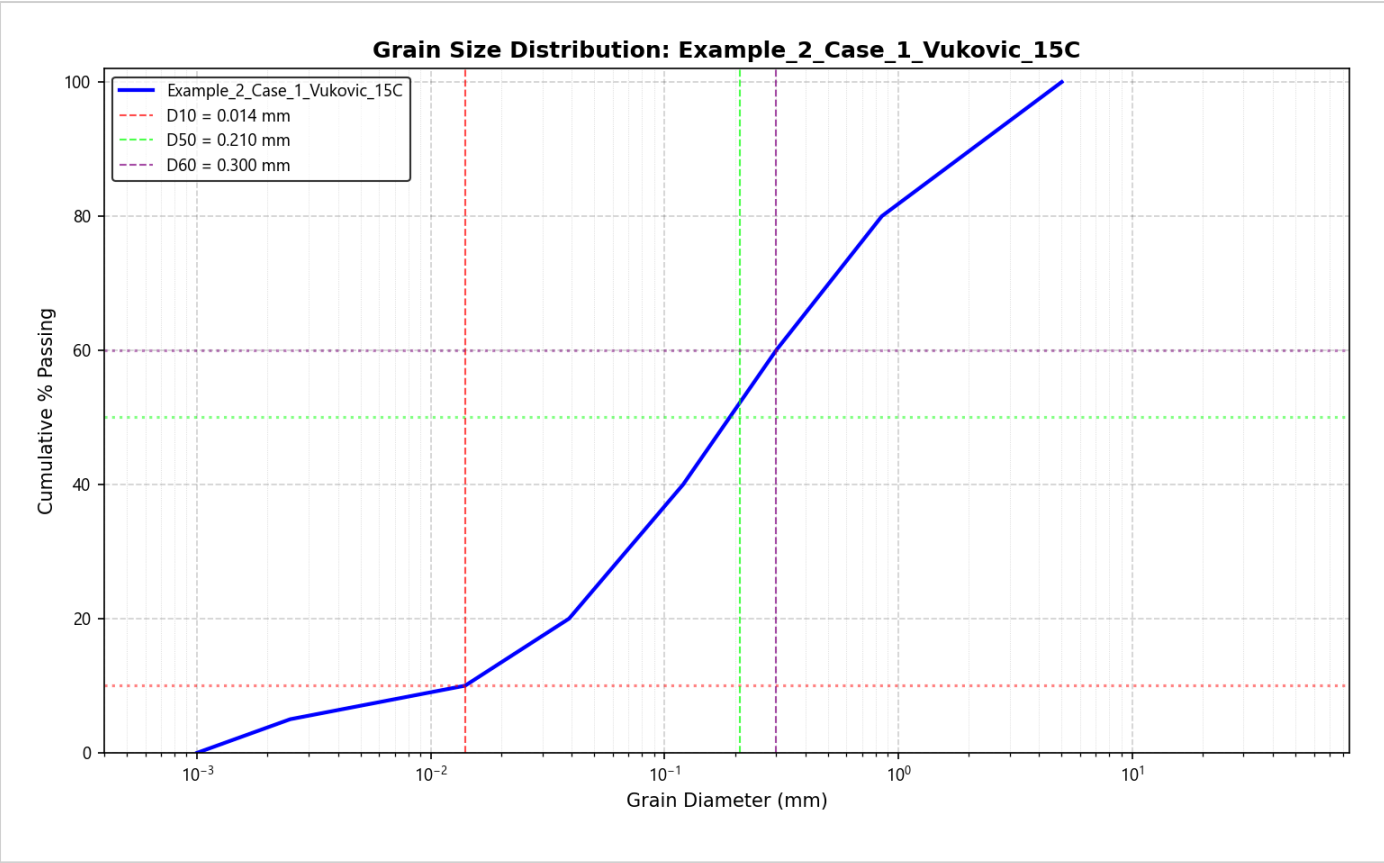
Hydraulic Conductivity Distribution

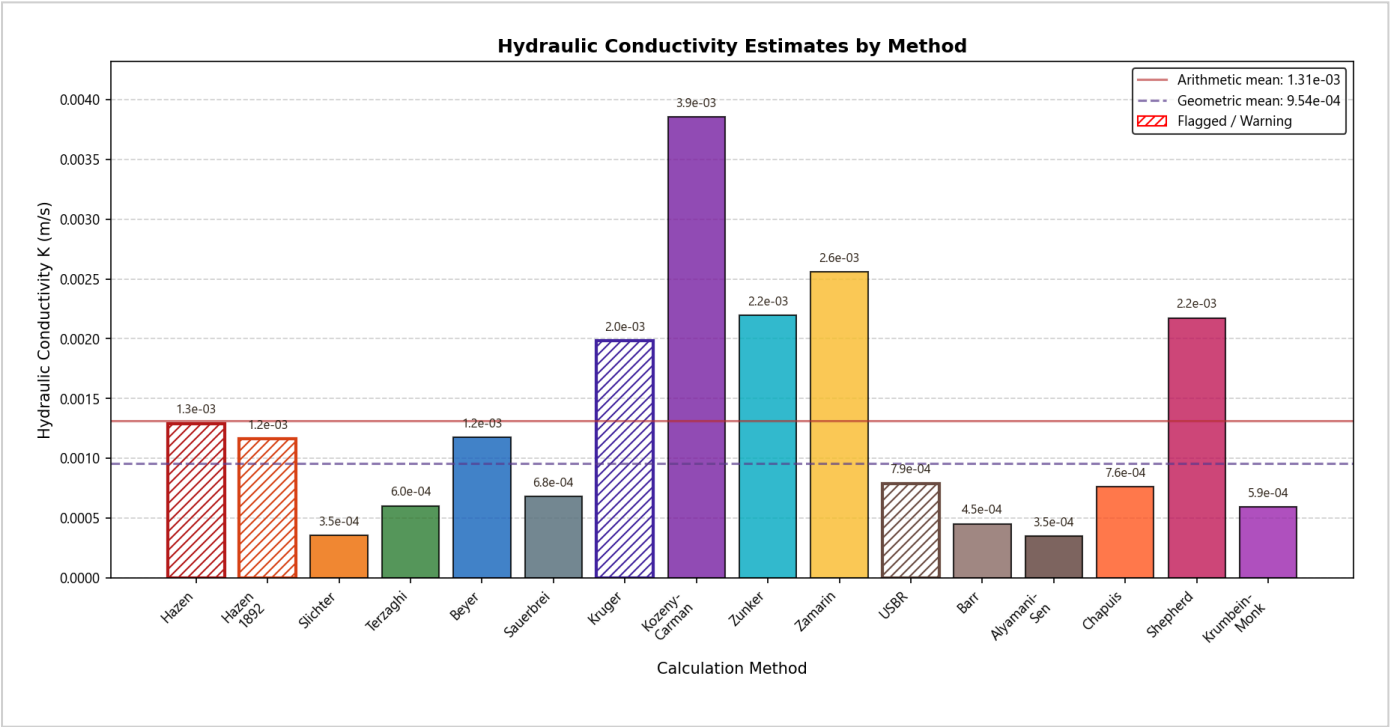
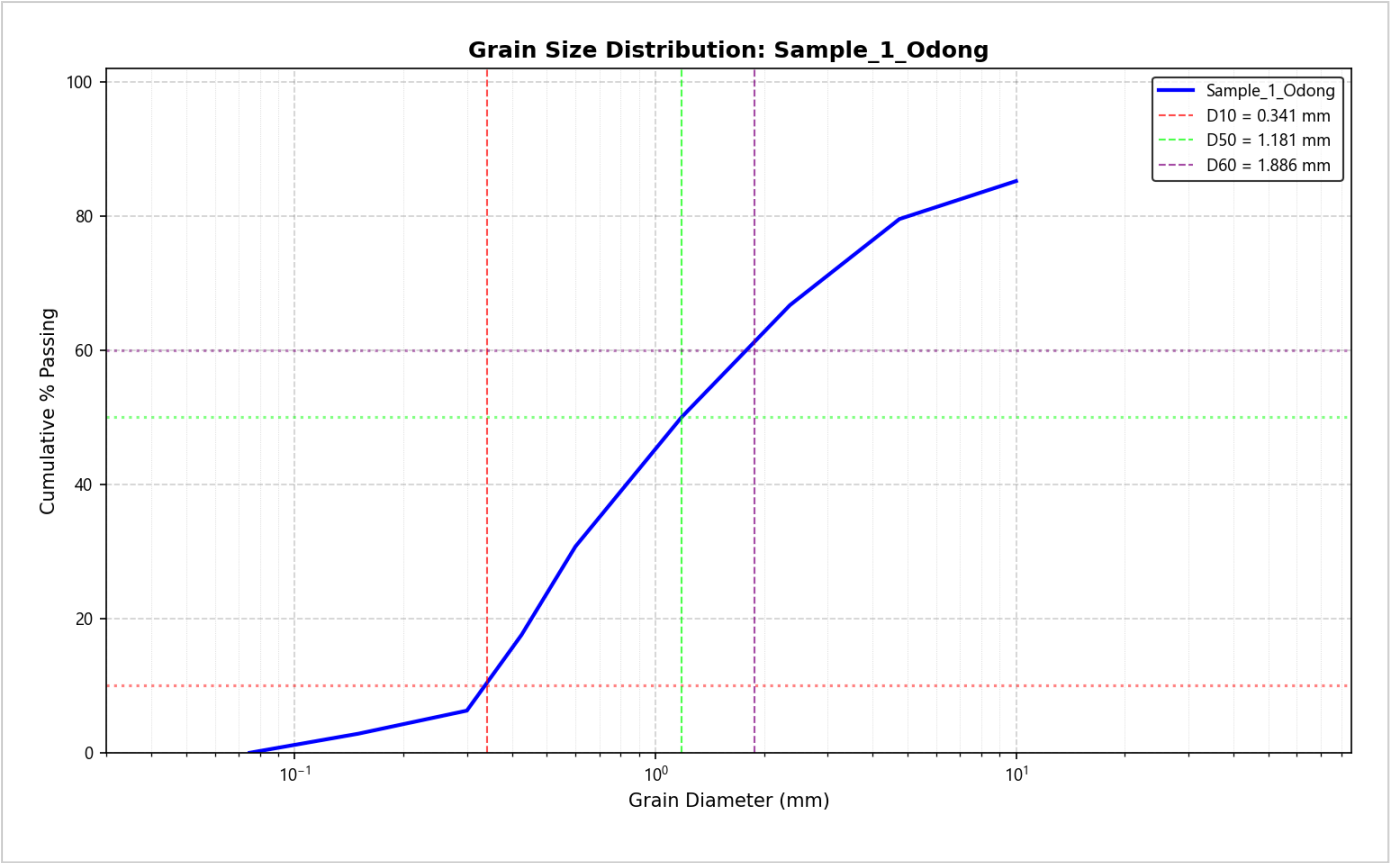


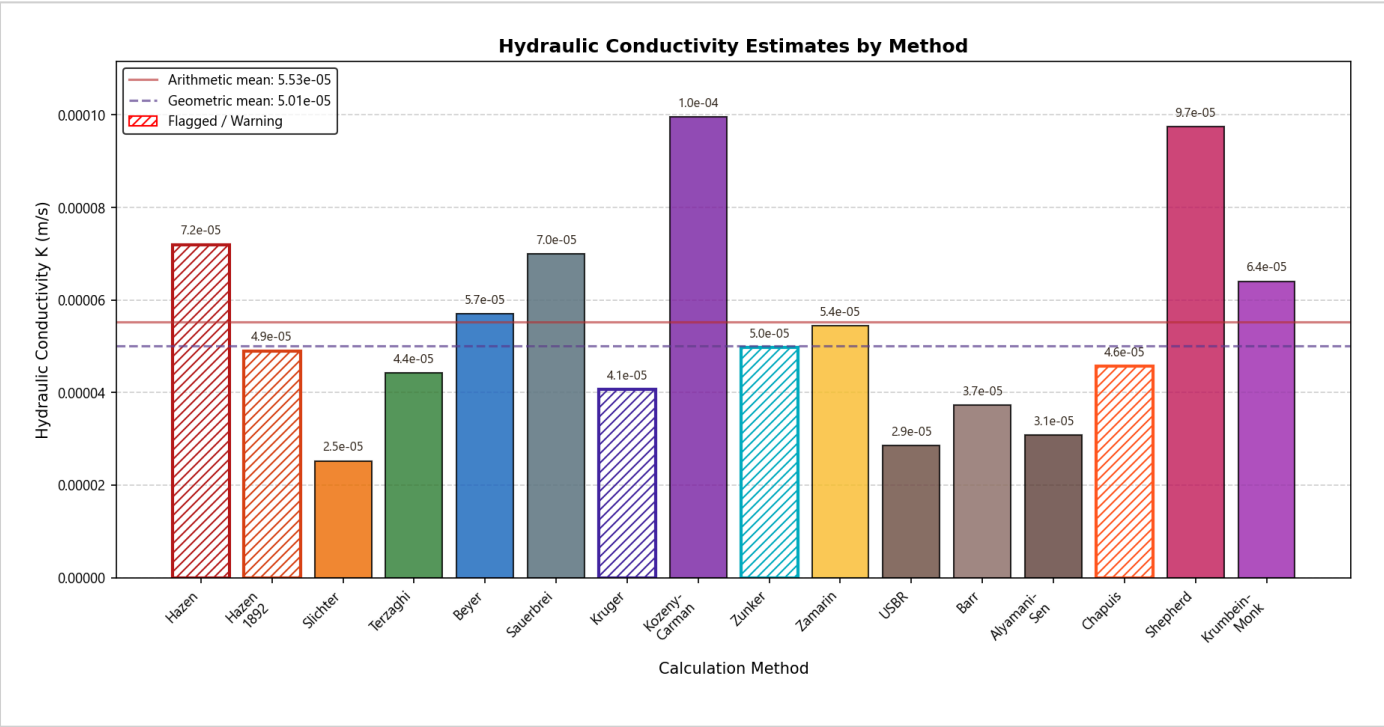
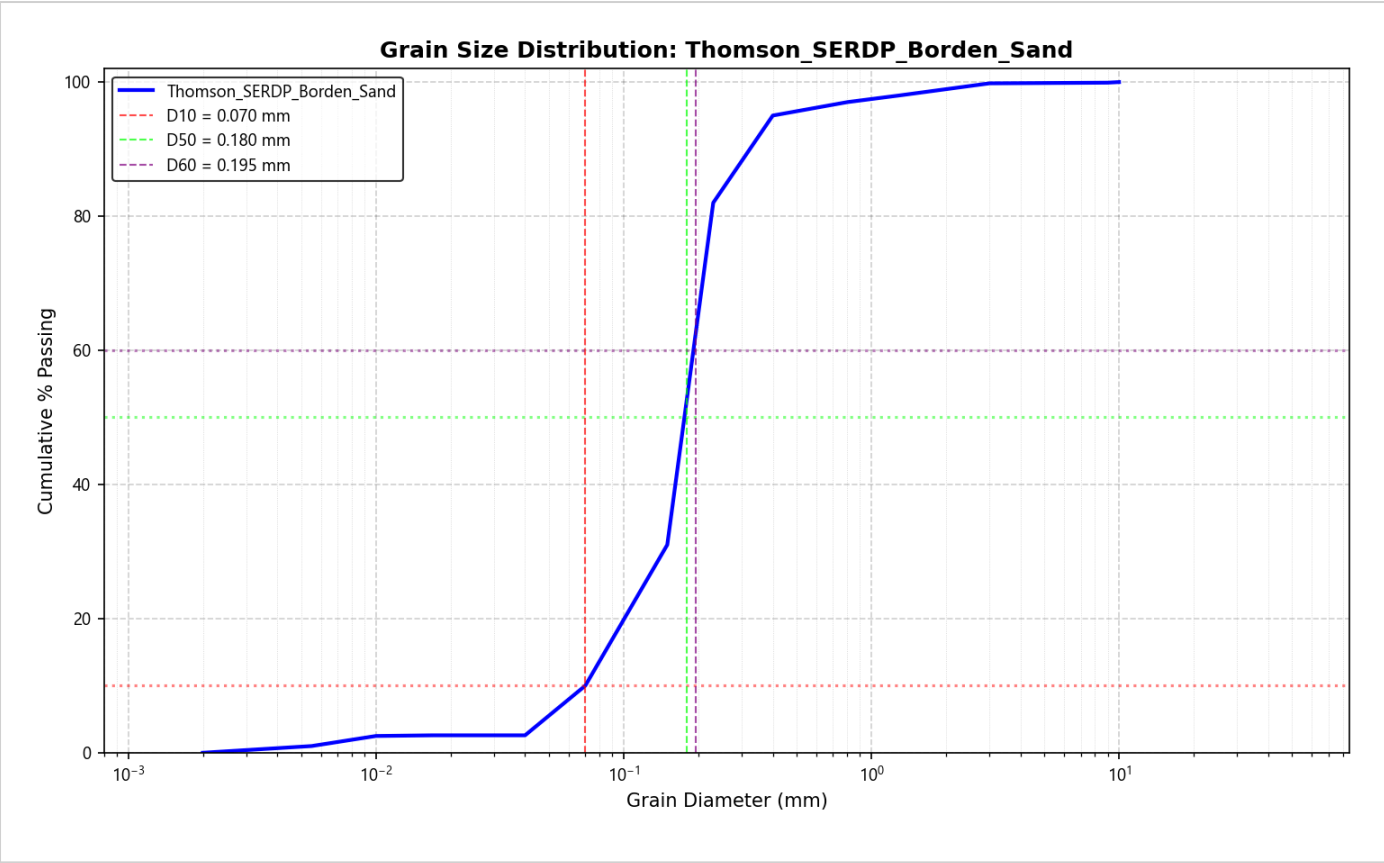
Example_1_Case_1_Vukovic

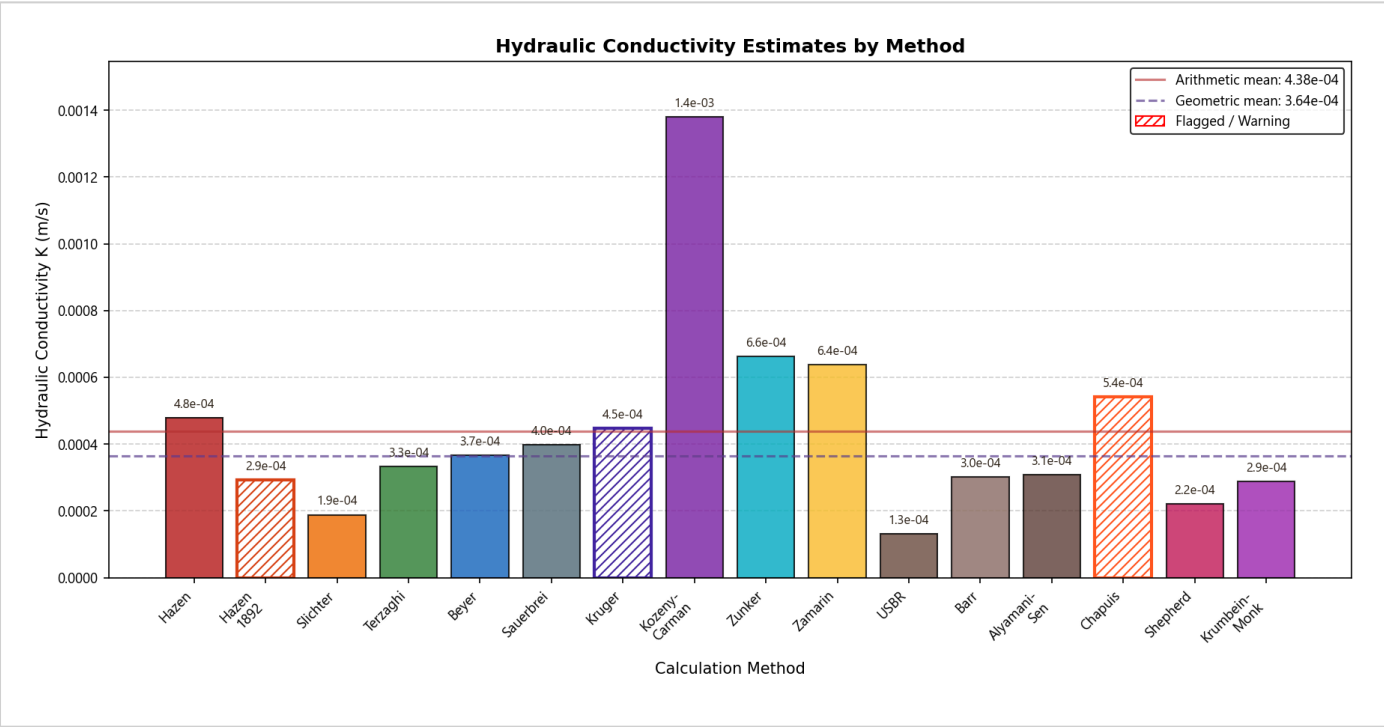
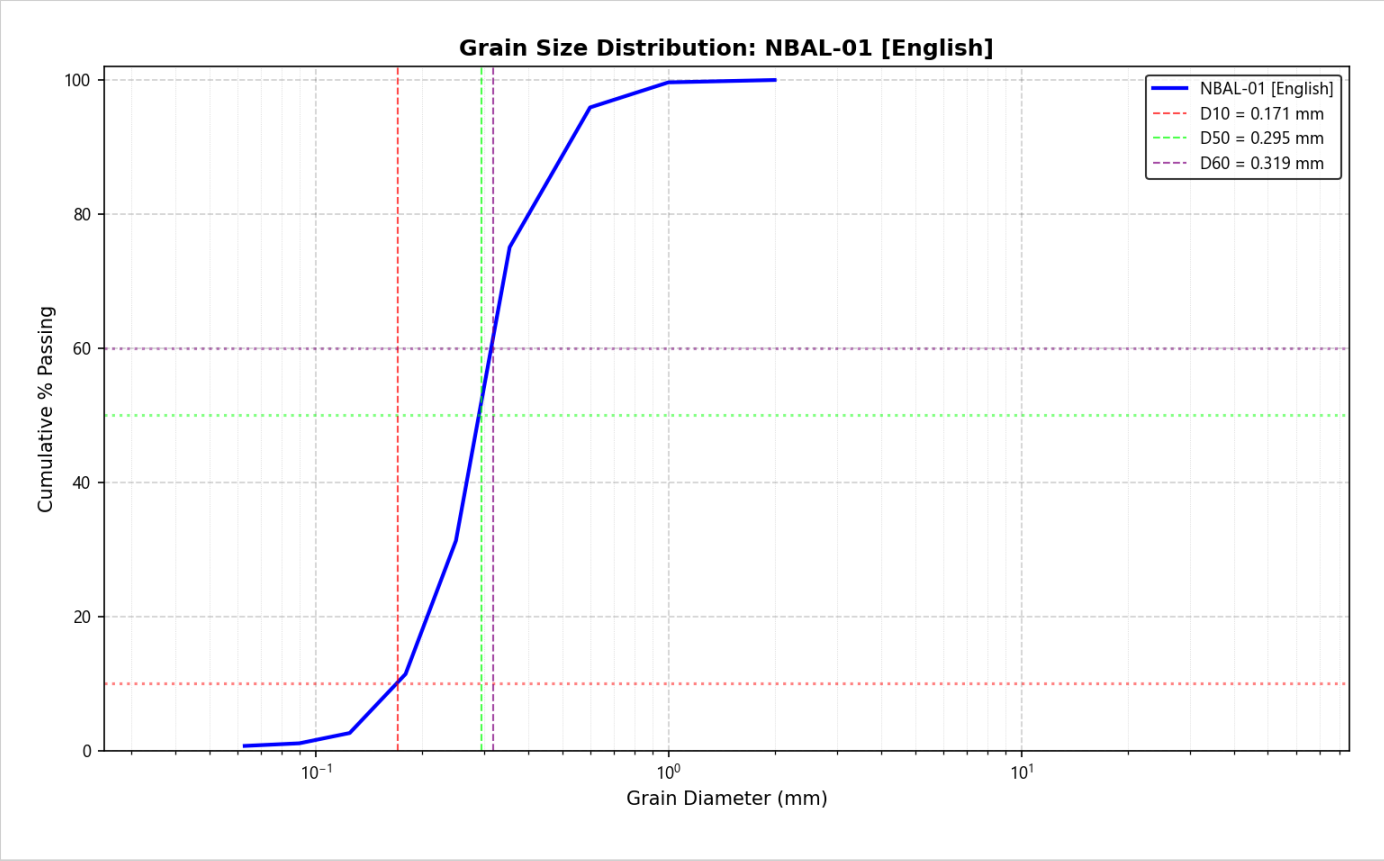


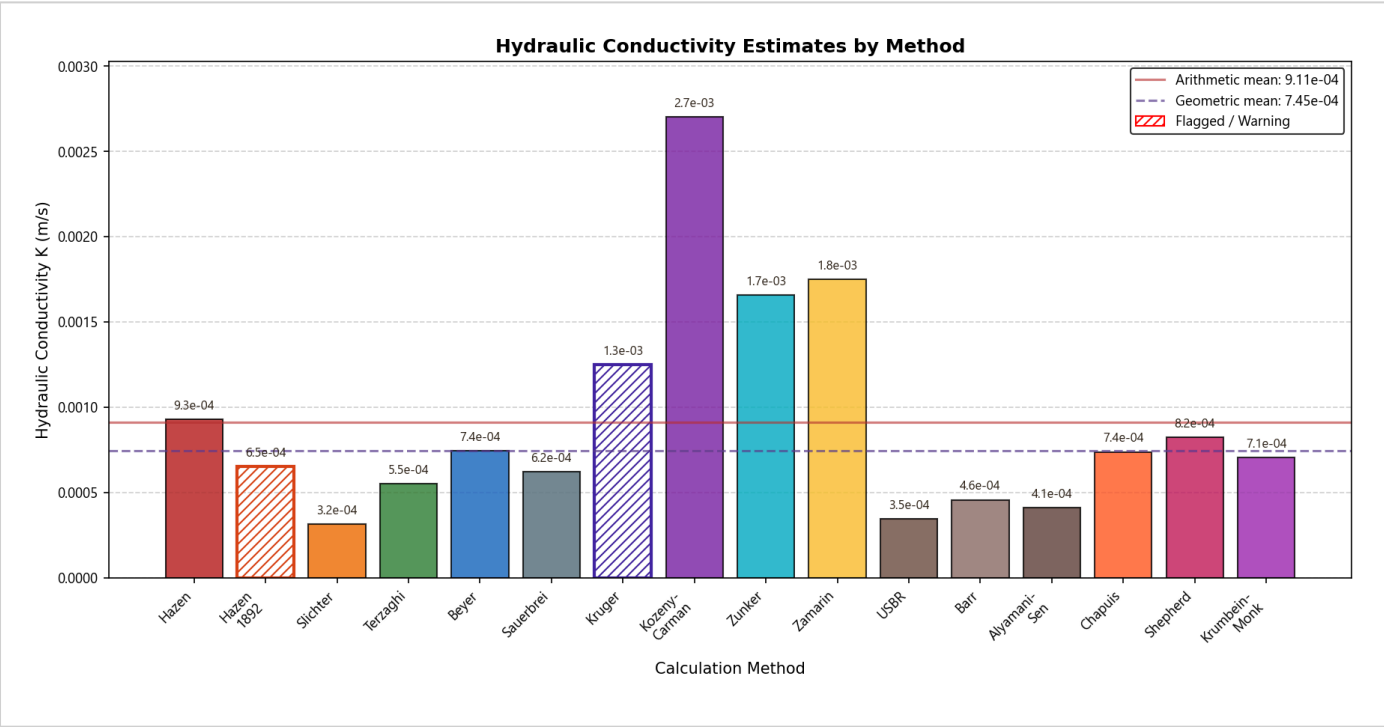
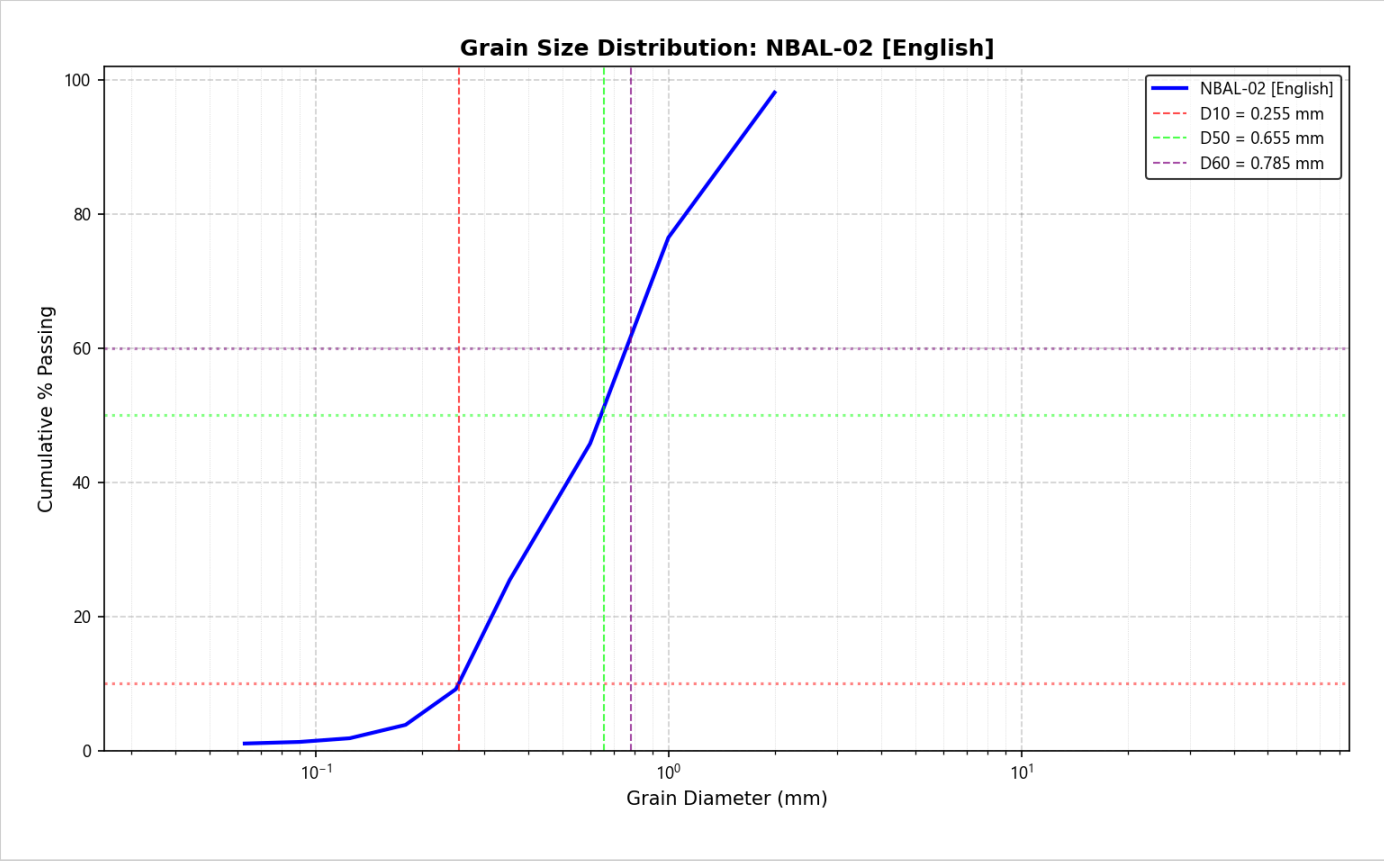


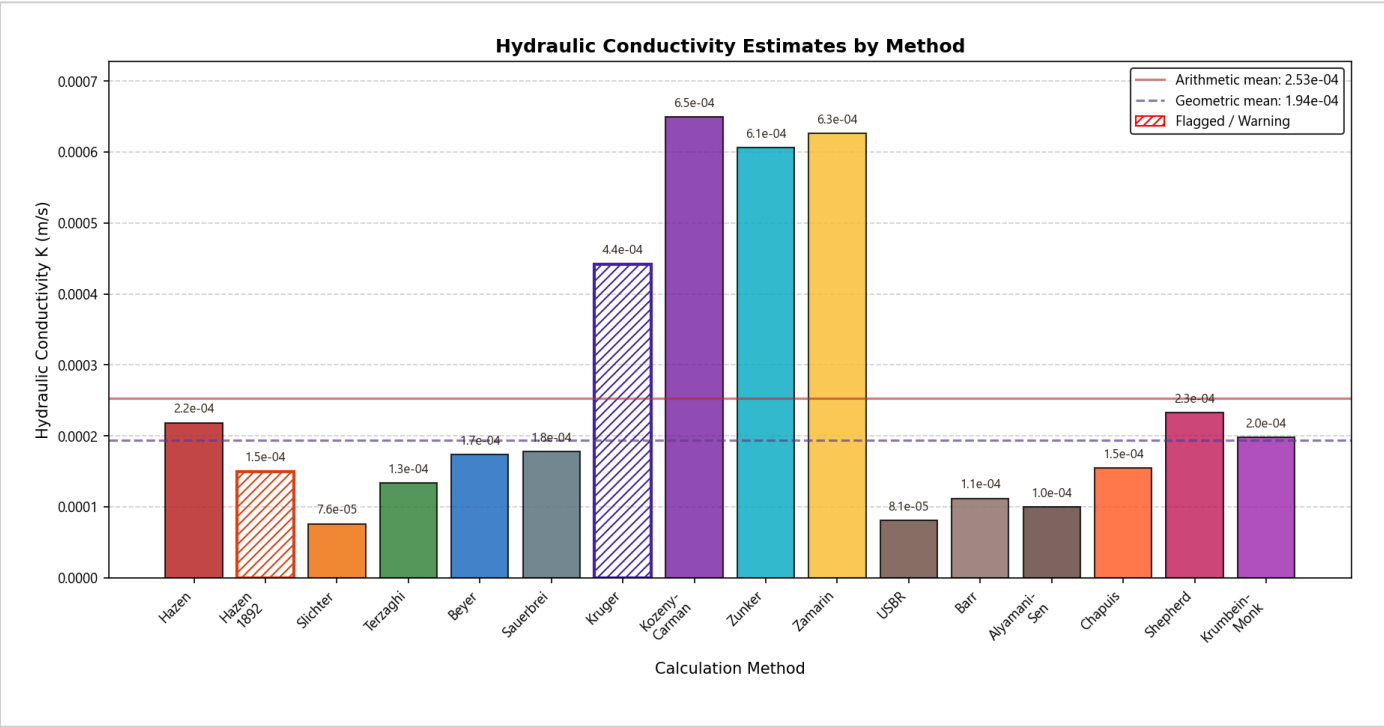
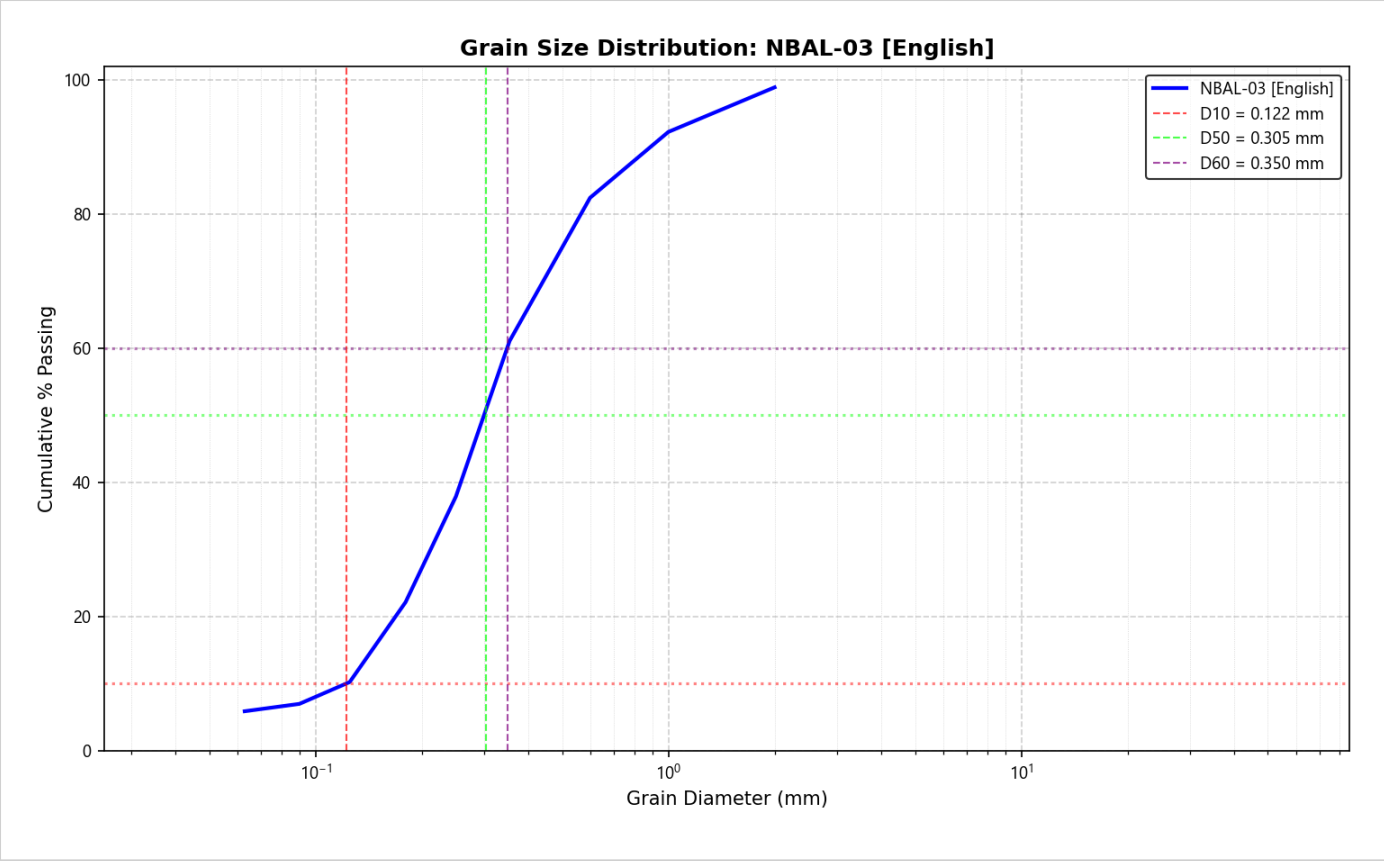


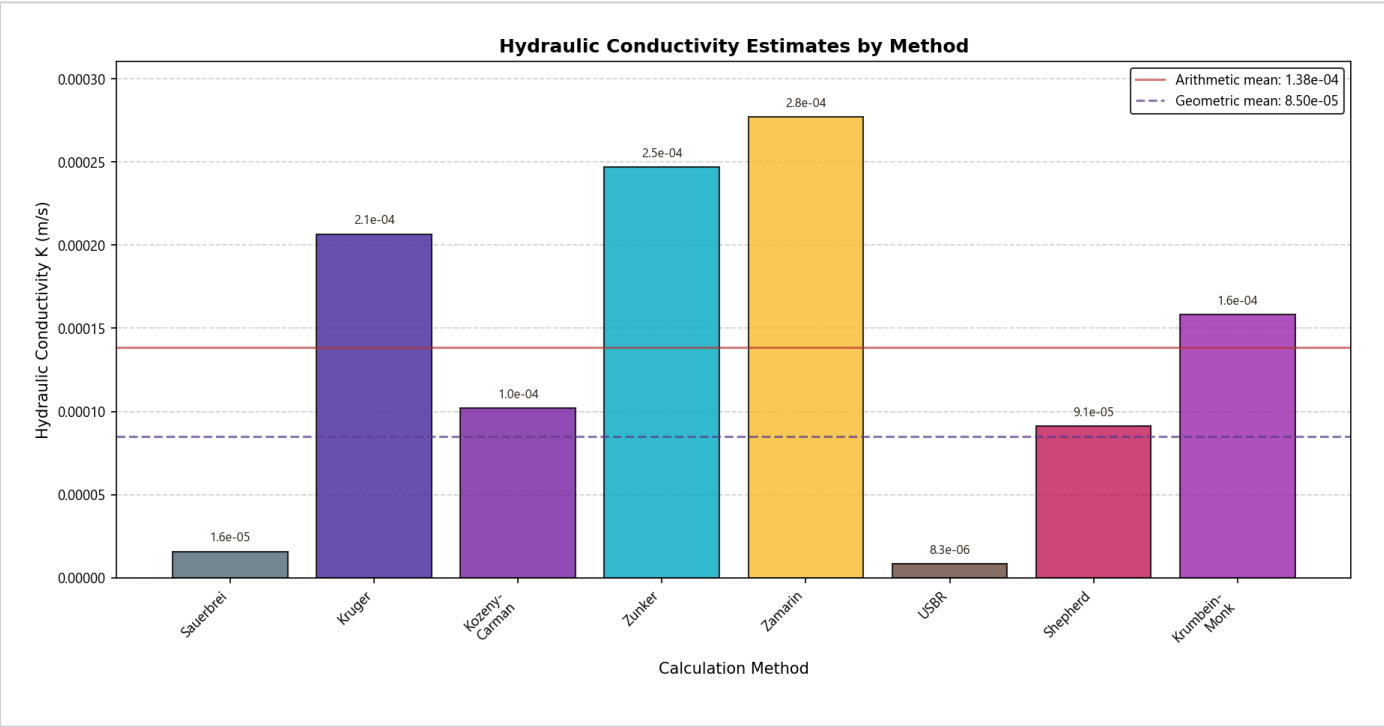
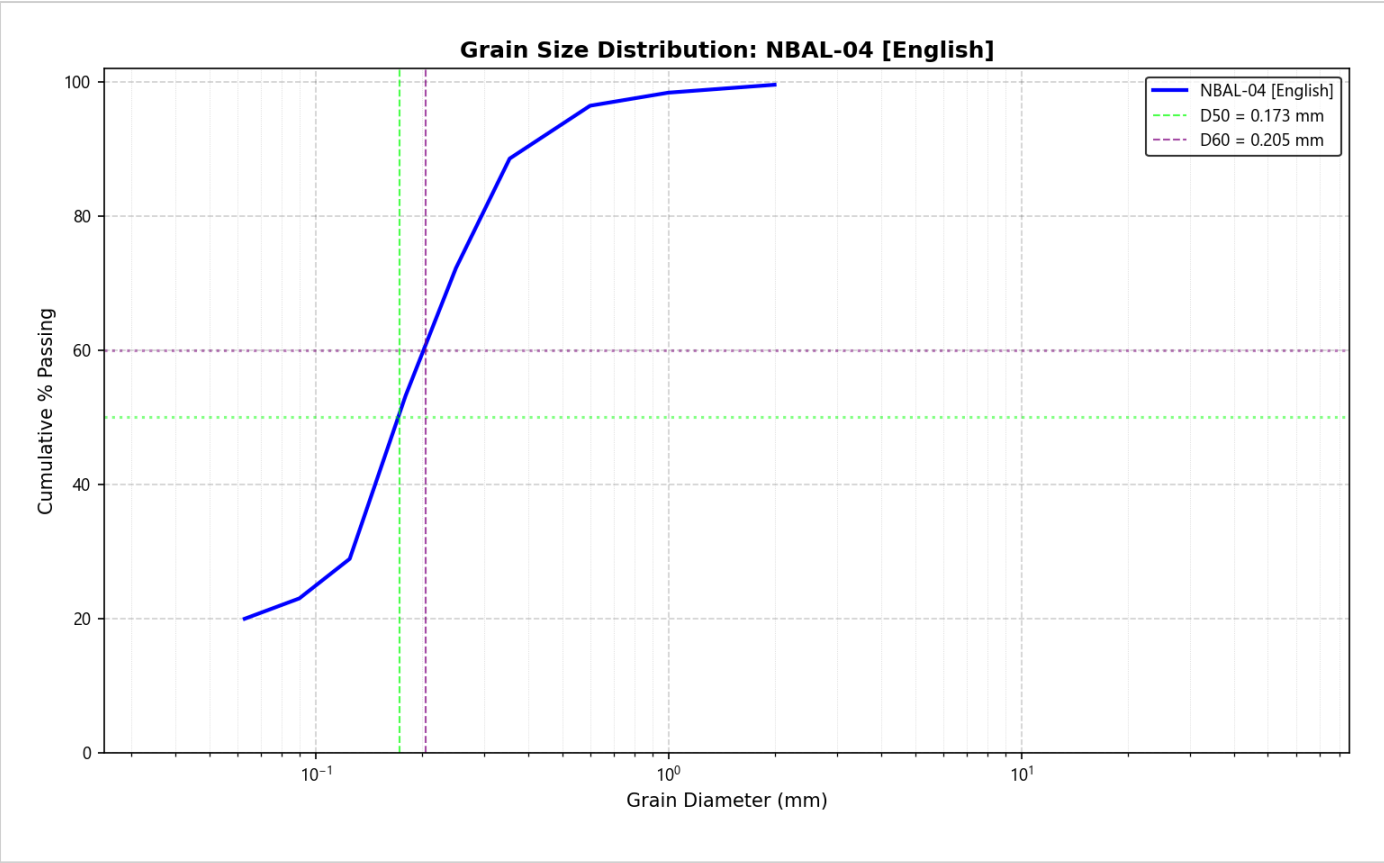


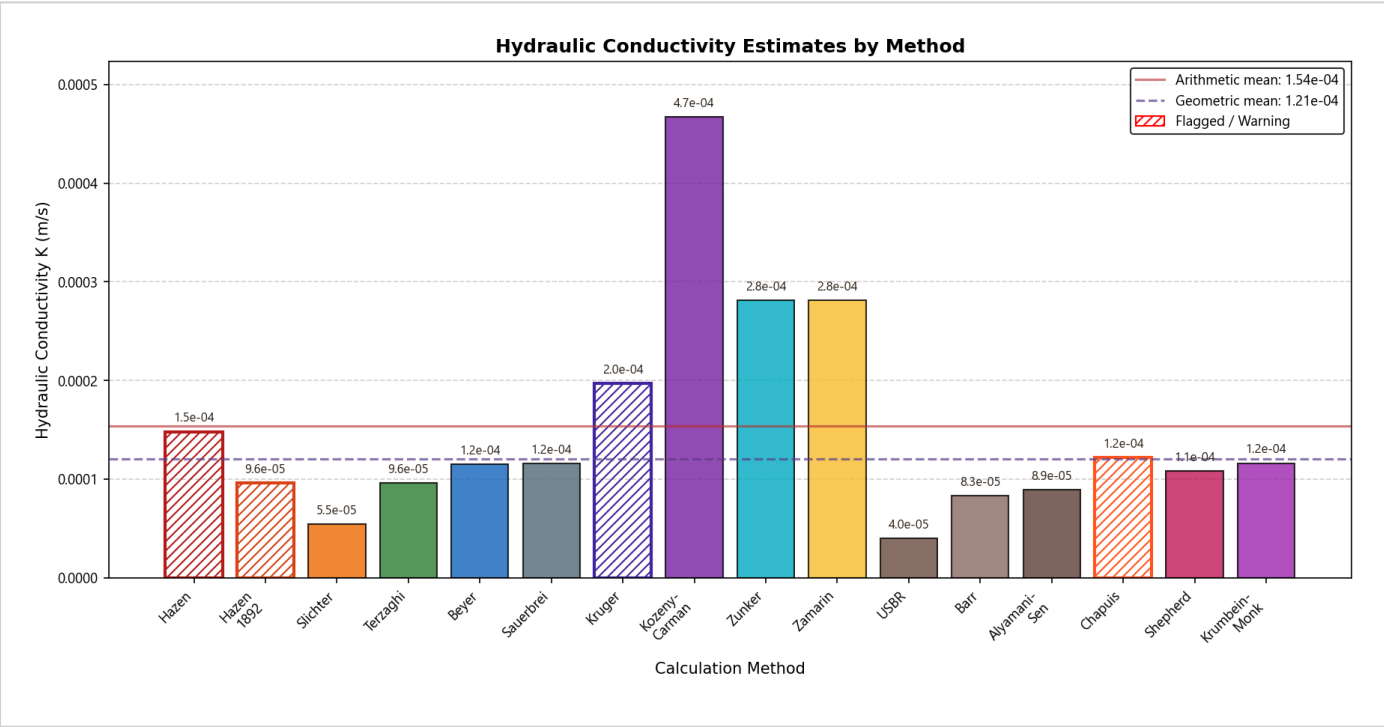
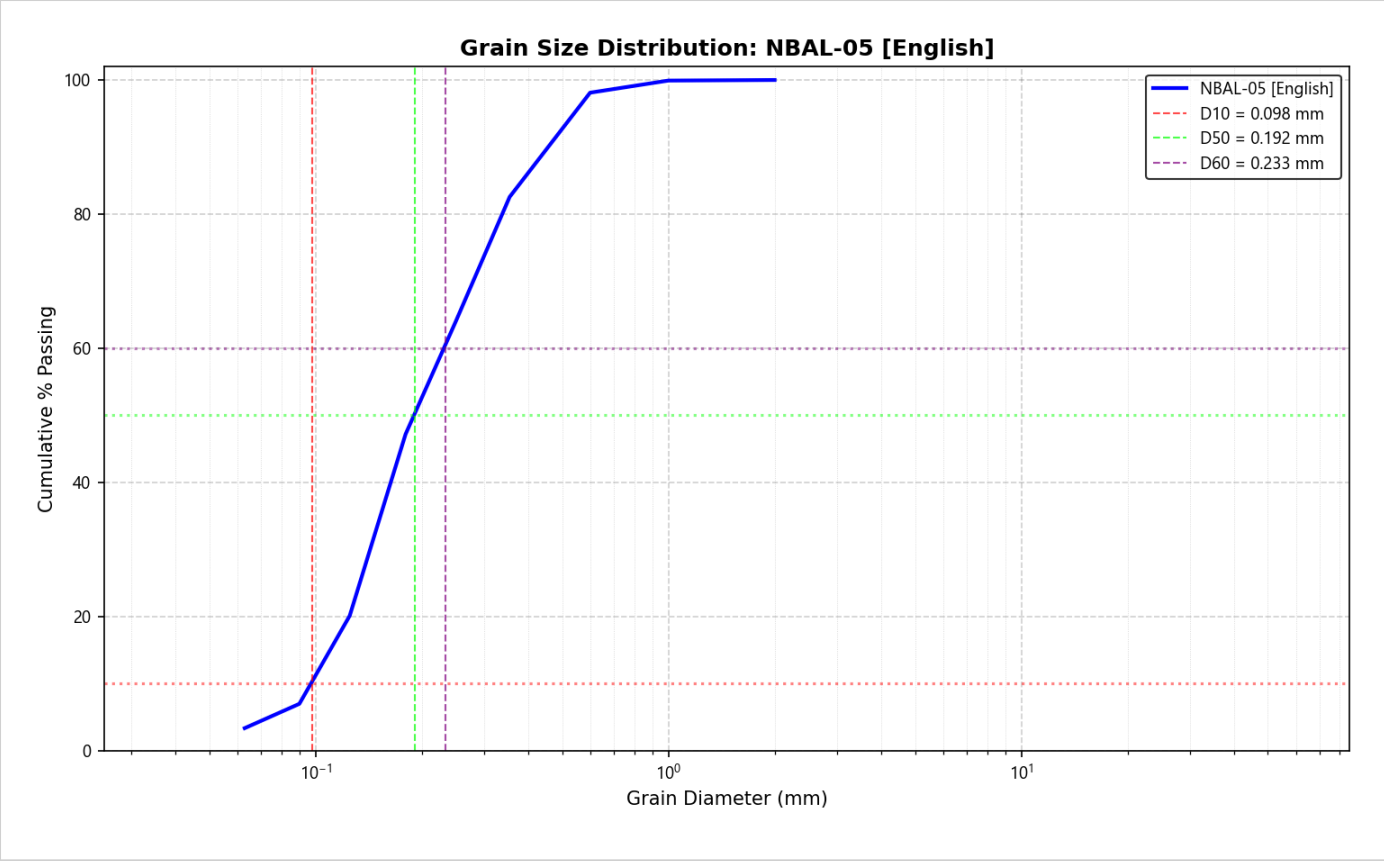


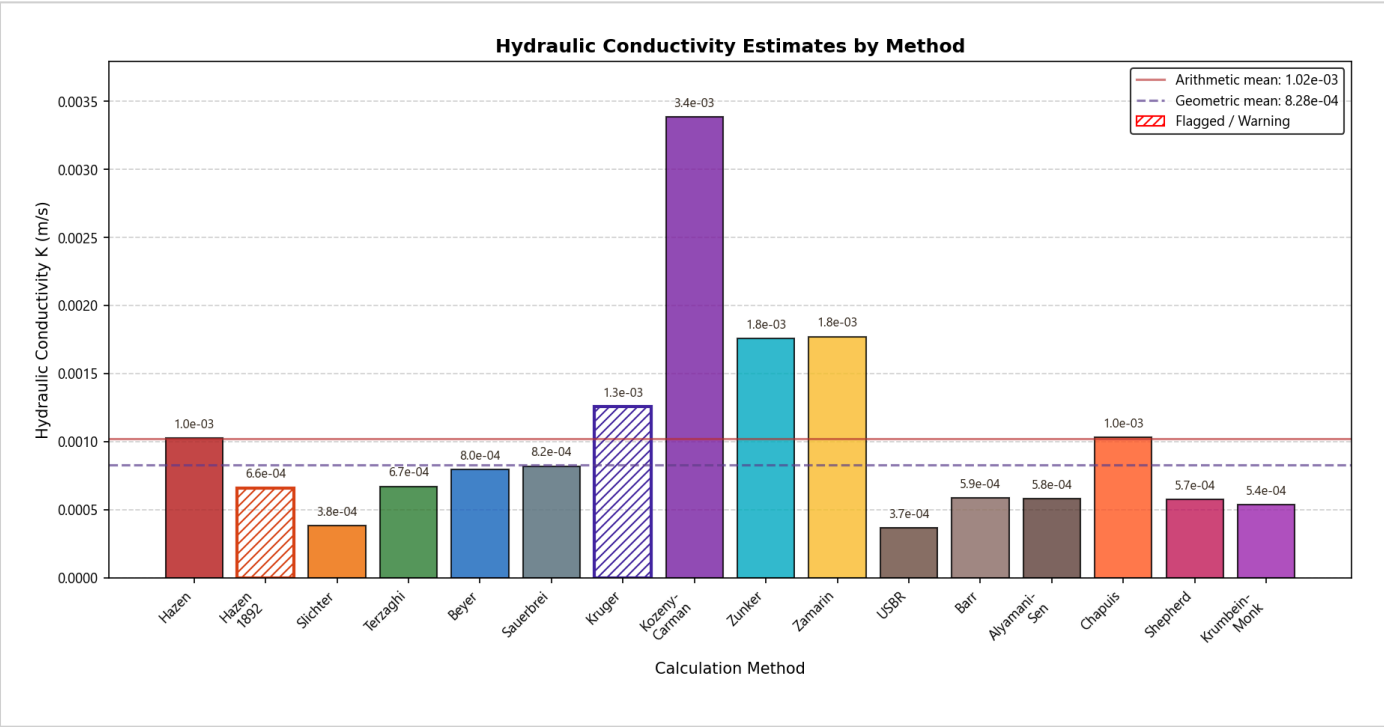
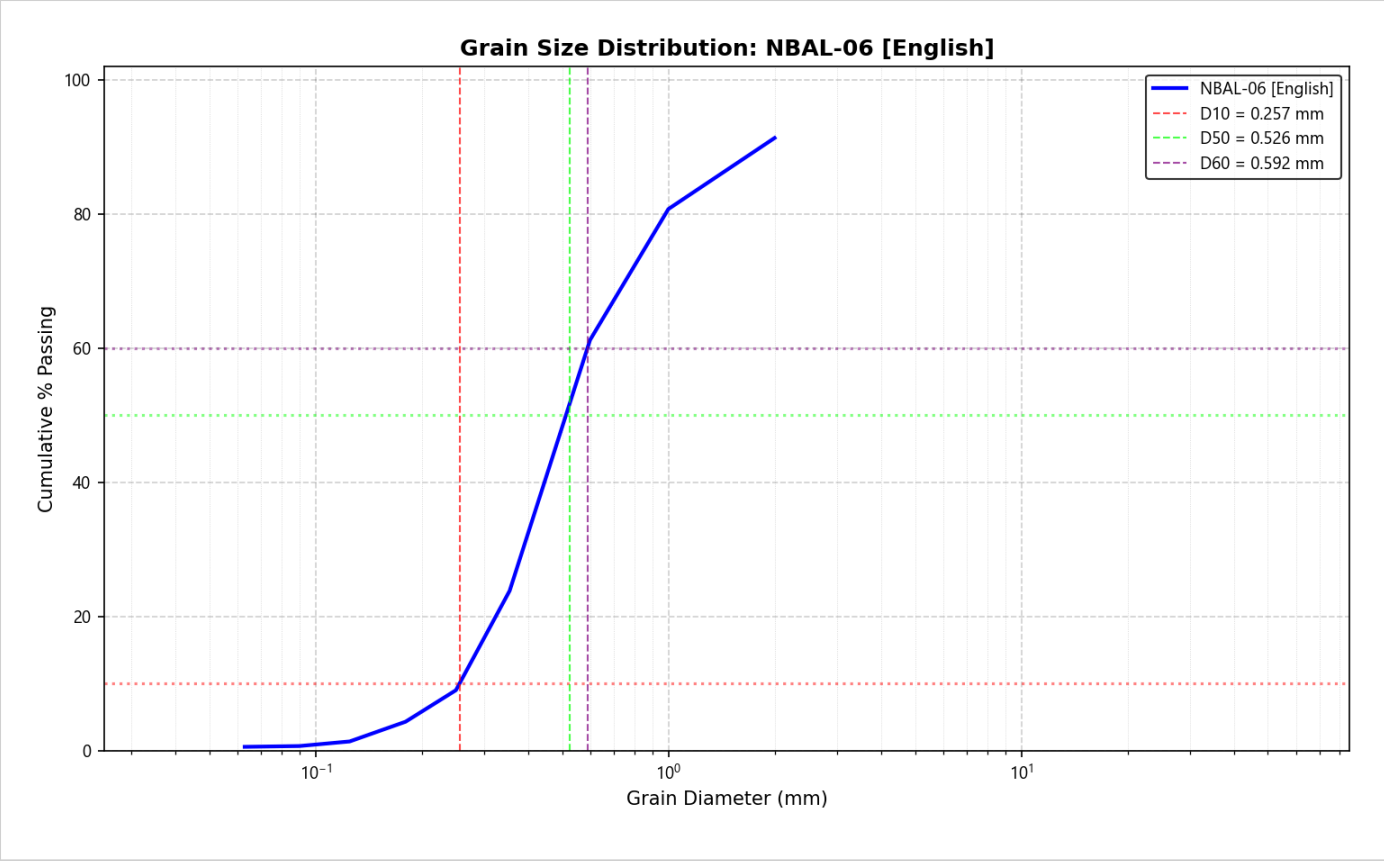


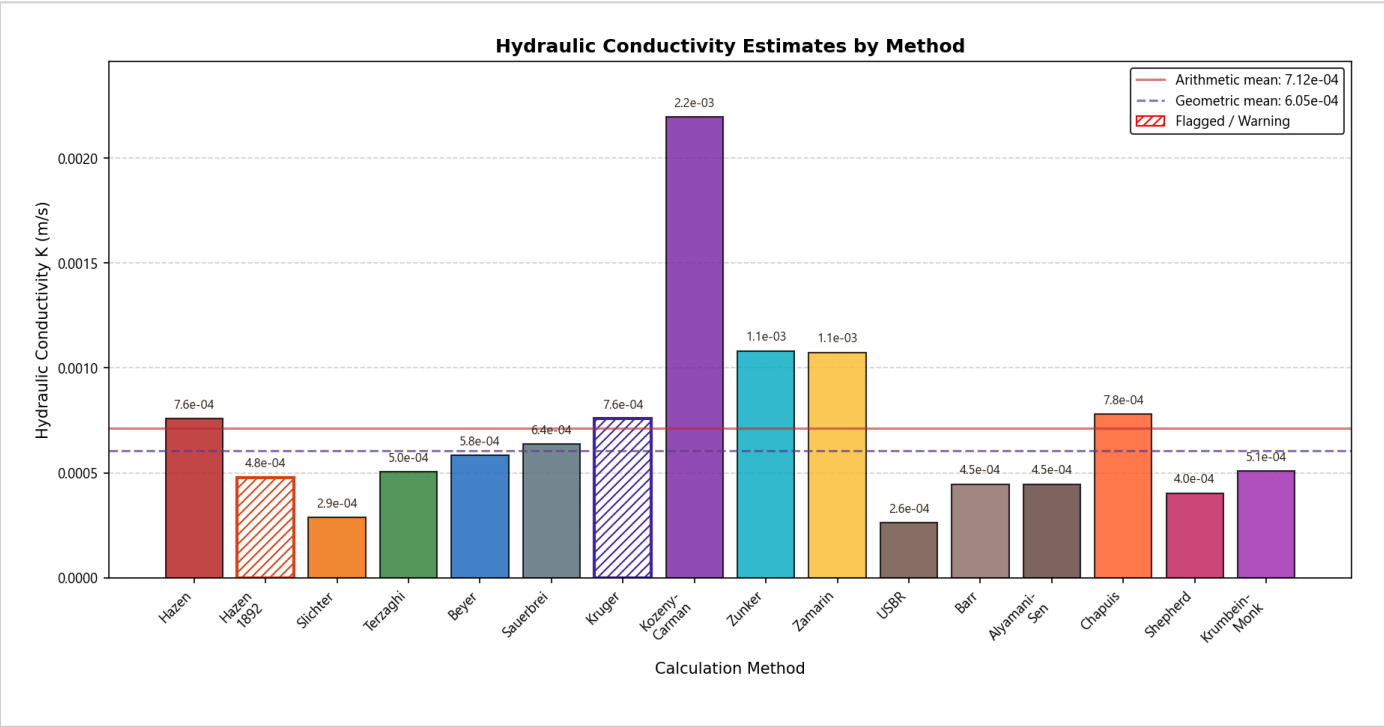
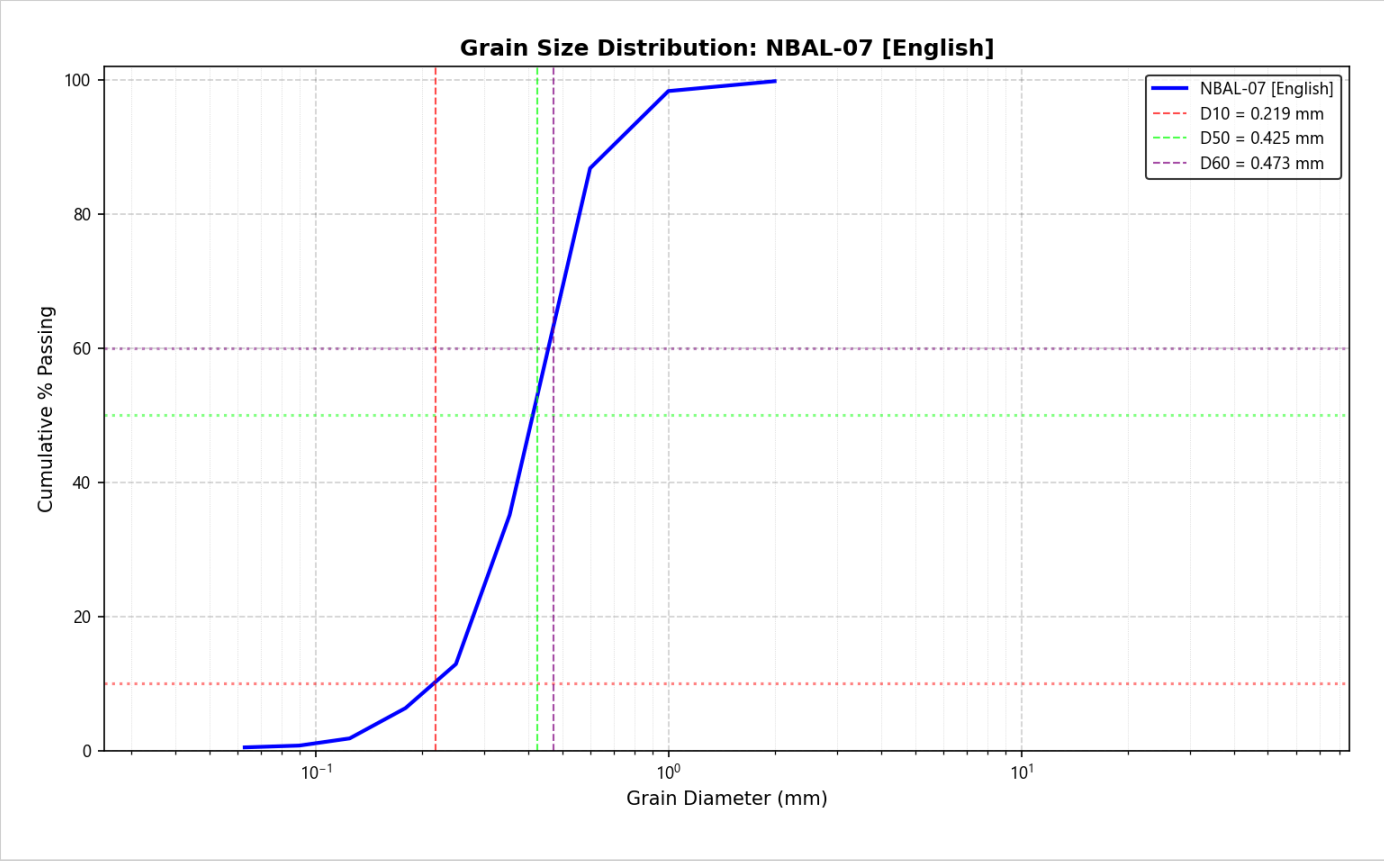


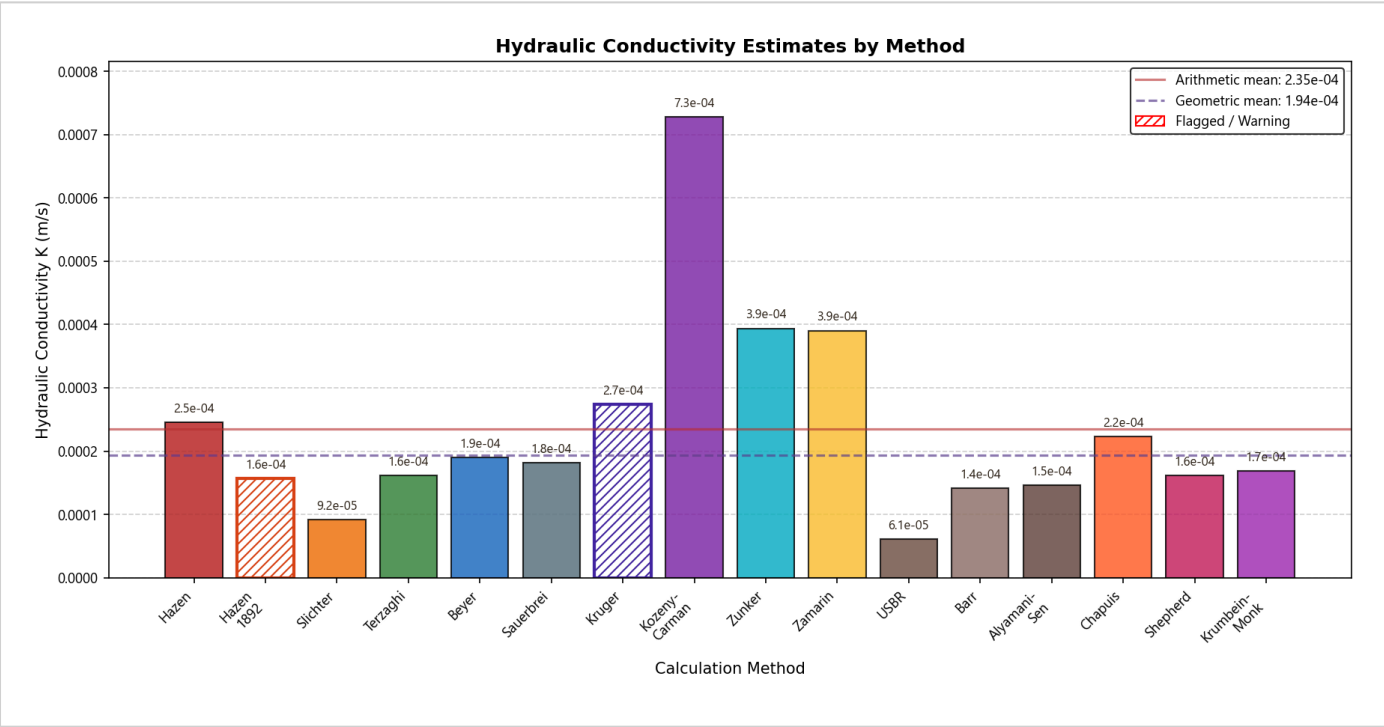
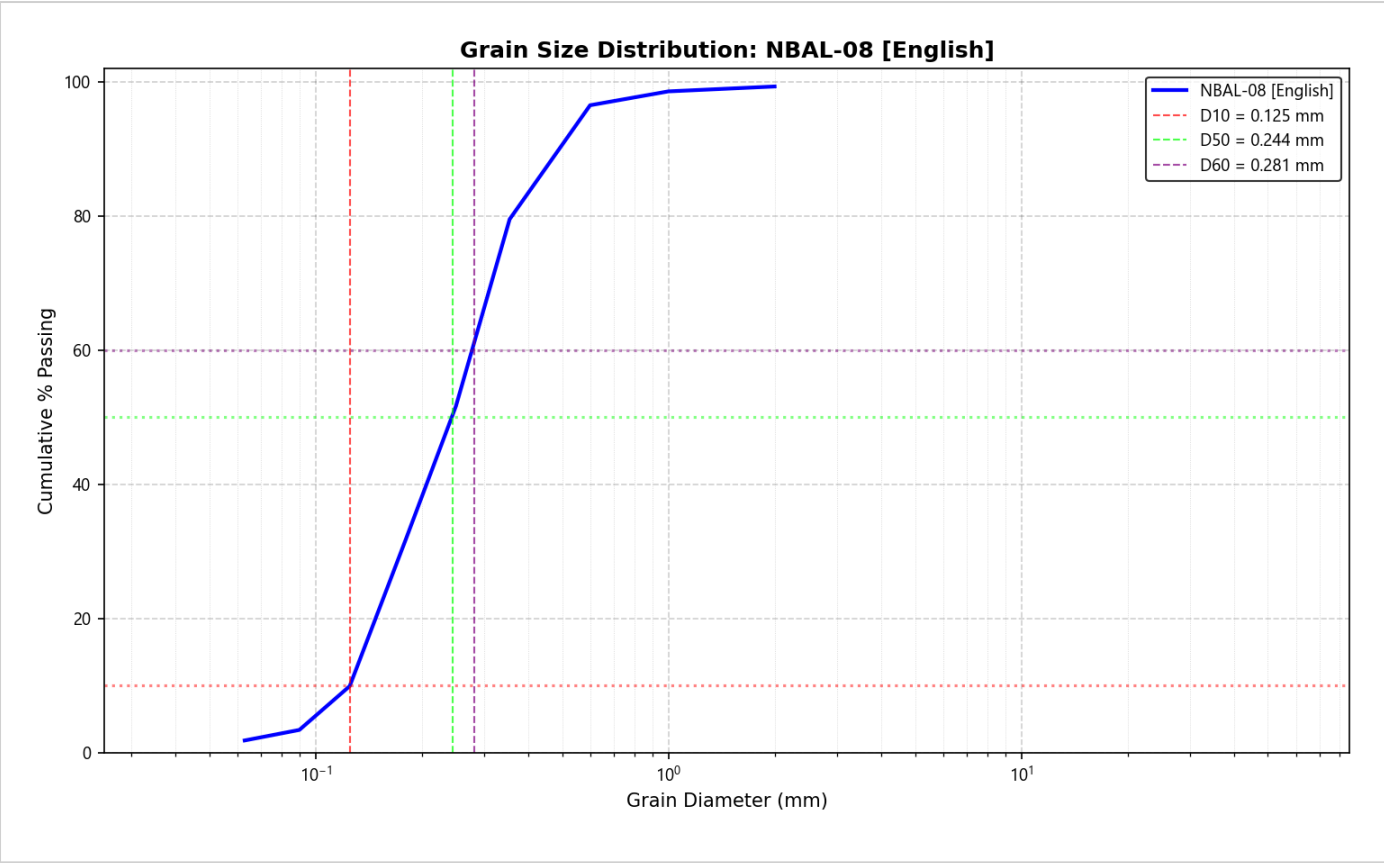


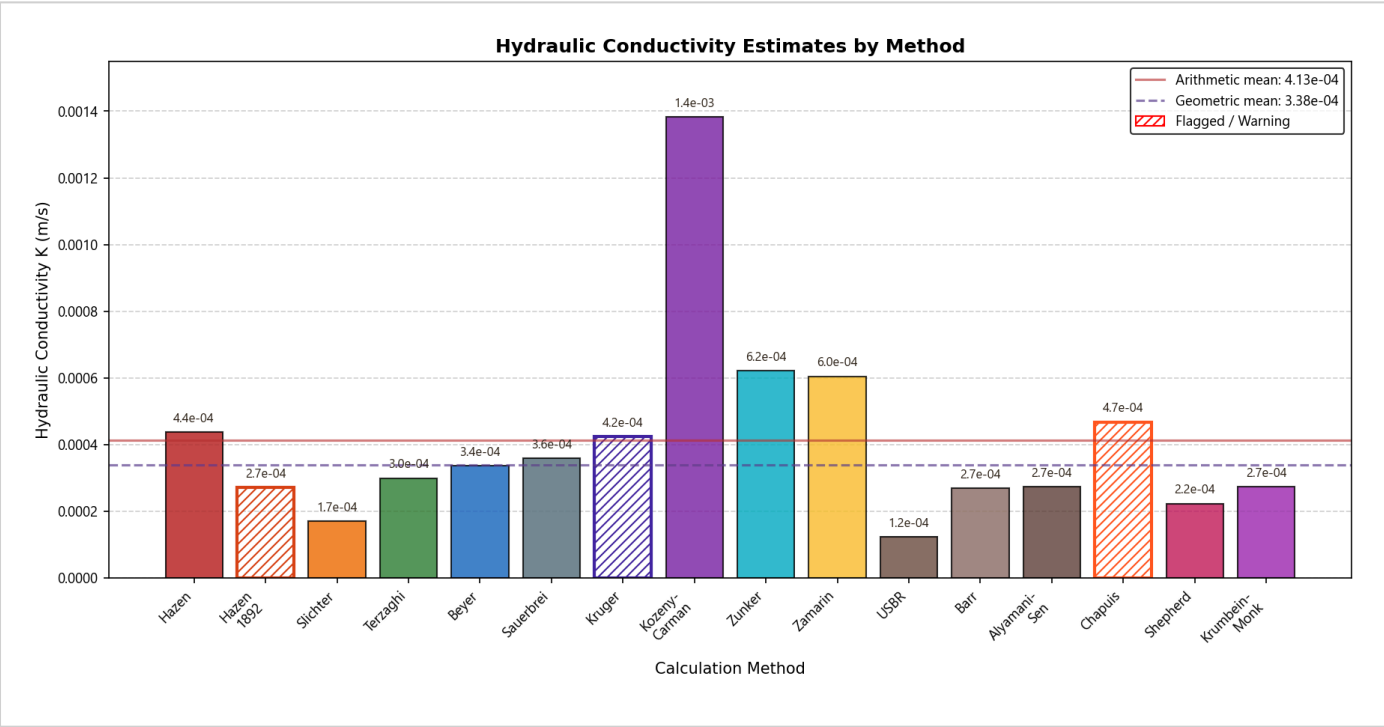
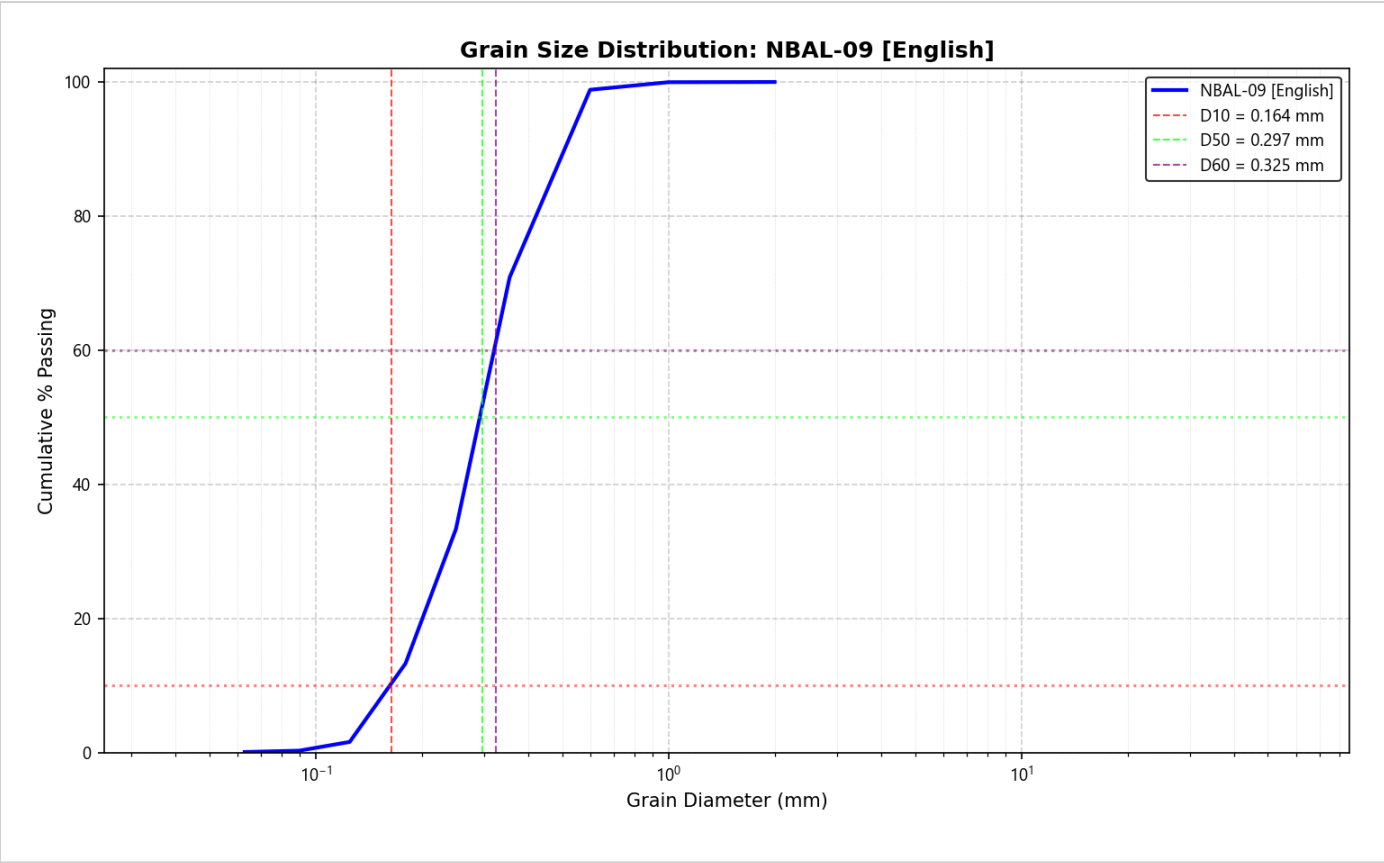


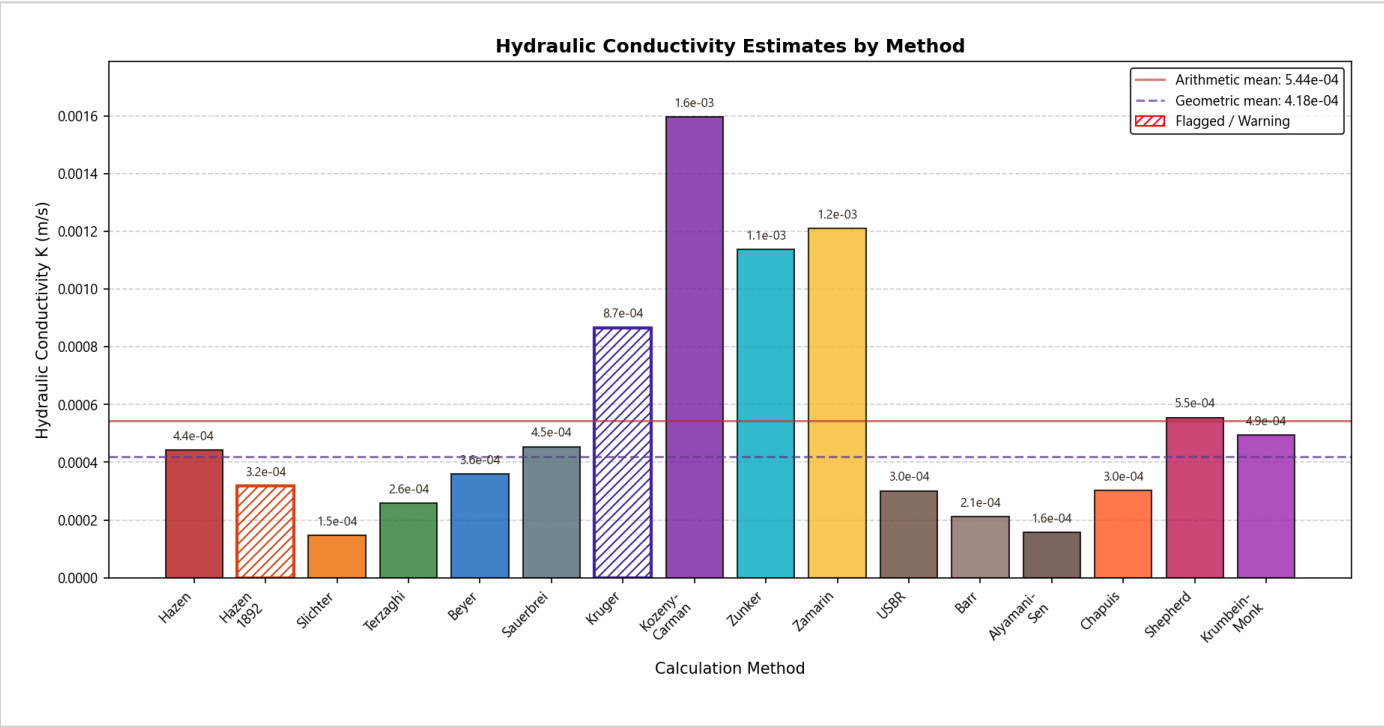
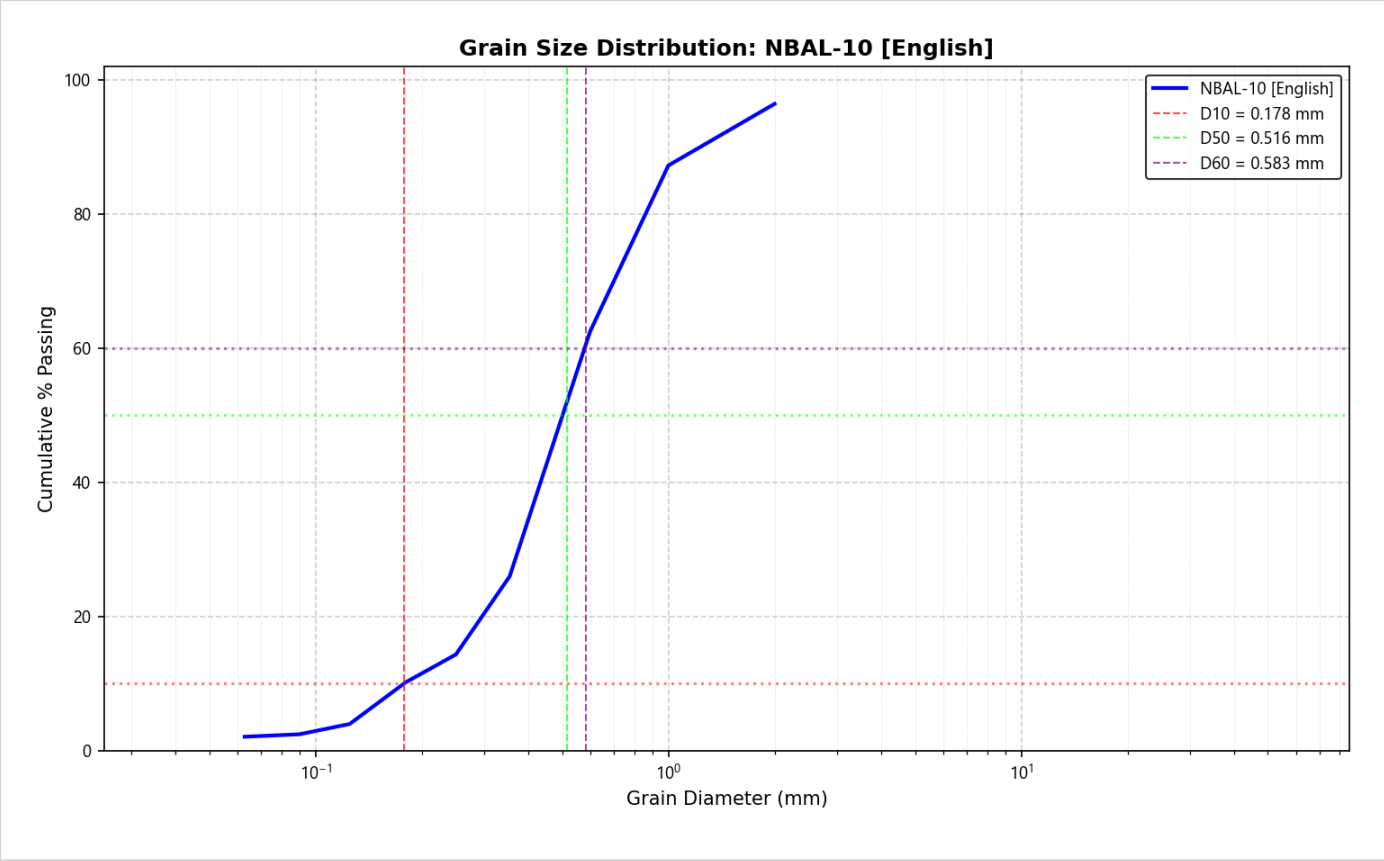












Interpretation & Discussion

Comparative Analysis

Permeability Characteristics:

- The highest permeability sample is Sample_1_Odong with $K = 9.54 \times 10^{-4}$ m/s, classified as High (Clean Sand).
- The lowest permeability sample is Example_2_Case_1_Vukovic_15C with $K = 1.14 \times 10^{-6}$ m/s, classified as Low (Silt).
- The 837.4-fold difference in permeability reflects the variability in grain size distribution among the samples.

Statistical Overview:

- Average sample K geometric mean: 3.63e-04 m/s
- Standard deviation: 2.84e-04 m/s
- Coefficient of variation: 78.3%